

North American Utilities

Back to School: In the Name of Load... What More in the Name of Load

North American Utilities

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Top Ten Utilities Themes:

1. **Where We Stand: Load Growth Tailwinds Accelerate**
2. **Favorite Names**
3. **Thoughts on Valuing TSR: P/E / (Y+G) Analysis**
4. **Rate Base vs EPS Growth & vs Capital Plan Rolls**
5. **VIU Generation Torque Analysis**
6. **Bill Pressure and O&M Analysis: Key Lever to Manage Bill Headroom**
7. **Regulators & Policy in Focus**
8. **Regulatory Risk Ranker**
9. **Multiple Expansion vs EPS Revision Analysis**
10. **Gas Views**

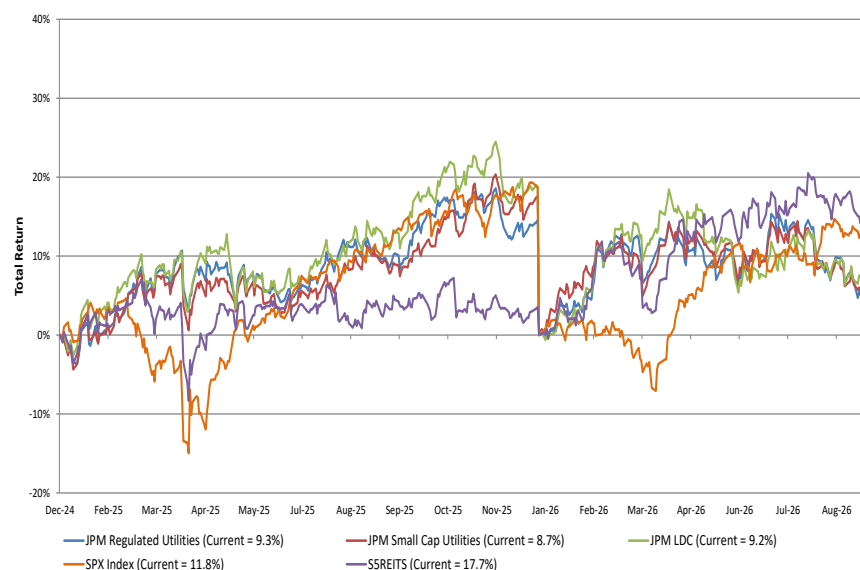
1) Where We Stand

Need for Nimbleness Given Surging Growth vs Affordability Concerns Amidst Busy Election Cycle

Robust Load-Driven Capex Backdrop Fuels Electric Growth

- Core utility value proposition = regulated rate base growth set to accelerate
 - Emerging data center-driven and onshoring load growth opportunities
 - Electrification, generation, transmission, reliability investments lifting capex
 - Favor IPPs, generation over wires, more torque to trends
 - Upside hurdle now resides with EPS increases, vs just capex
- We see a heavy 2026 election cycle accentuating the main 2025 debate: data center load growth tailwinds vs affordability concerns
 - Companies that can demonstrate moderating or freezing ratepayer bill inflation stand as best positioned during politically charged year
 - O&M management = key controllable utility input to manage bills
- Energy usage and local zoning resonate as key considerations for new load
 - Location matters: states welcoming (southeast) vs not
 - Demand response pricing continues to evolve for capacity needs
 - Data center deals ultimately remain beholden to local community support

Returns by Different Sectors



Coverage Universe Total Return by Subsector

	Total Return (%)																	
Symbol	2020	2021	2022	2023	1Q24	2Q24	3Q24	4Q24	2024	1Q25	2Q25	3Q25	4Q25	2025	1Q26	2Q26	3Q26 YTD	2026 YTD
Relevant Indices																		
SPX	18.4%	28.7%	-18.1%	26.3%	10.6%	4.3%	5.9%	2.4%	25.0%	-4.3%	10.9%	8.1%	2.7%	17.9%	-4.4%	15.2%	1.4%	11.8%
SPX Energy	-33.7%	54.4%	65.5%	-1.4%	13.7%	-2.4%	-2.3%	-2.4%	5.7%	10.2%	-8.6%	6.2%	1.5%	8.7%	38.2%	-13.5%	23.1%	47.2%
SPX Utilities	0.5%	17.7%	1.6%	-7.1%	4.6%	4.7%	19.4%	-5.5%	23.4%	4.9%	4.3%	7.6%	-1.4%	16.0%	8.3%	-0.5%	-5.8%	1.4%
UTY	2.7%	18.2%	0.6%	-9.2%	5.4%	4.5%	19.4%	-8.1%	20.9%	7.5%	2.0%	7.3%	-0.5%	17.1%	8.8%	-0.7%	-4.4%	3.4%
Regulated Utilities	-0.3%	12.9%	2.5%	-5.2%	0.8%	1.4%	19.6%	-4.9%	15.5%	12.6%	-3.6%	9.3%	-3.6%	13.9%	11.6%	0.0%	-6.5%	3.6%
Diversified Utilities	-3.9%	13.6%	-5.1%	-14.0%	5.8%	0.8%	16.6%	-5.1%	16.8%	0.0%	0.0%	7.6%	-0.6%	13.9%	10.8%	1.1%	-6.0%	3.8%
LDC Utilities	-16.7%	19.4%	9.6%	0.4%	3.3%	0.2%	12.1%	1.0%	15.7%	9.9%	0.3%	8.5%	-1.3%	16.9%	10.8%	2.5%	-3.8%	1.2%
Canadian Utilities	1.3%	21.8%	-14.3%	2.5%	-0.7%	-2.8%	16.7%	-1.8%	13.3%	14.3%	4.1%	8.2%	2.4%	31.9%	7.8%	5.4%	-8.1%	4.4%
Water Utilities	13.2%	16.2%	-2.2%	-14.7%	-9.8%	1.4%	10.9%	-14.5%	-13.0%	15.6%	-4.7%	-2.4%	-2.1%	5.2%	12.9%	0.9%	5.7%	20.5%
IPPs	-7.0%	19.1%	5.1%	69.2%	52.9%	16.7%	34.1%	6.3%	153.8%	-5.3%	62.8%	1.6%	-6.7%	68.8%	-11.4%	3.0%	-13.1%	-17.8%
Average	-2.5%	22.2%	4.5%	4.7%	8.6%	2.9%	15.2%	-3.3%	27.7%	6.6%	6.8%	6.2%	-1.0%	21.0%	9.3%	1.3%	-1.7%	7.9%
Median	0.1%	18.6%	1.1%	-3.3%	5.0%	1.4%	16.7%	-3.7%	16.2%	8.7%	1.1%	7.6%	-1.0%	16.5%	9.8%	1.0%	-5.1%	3.7%

Source: Bloomberg Finance L.P. Priced 9/10/2026

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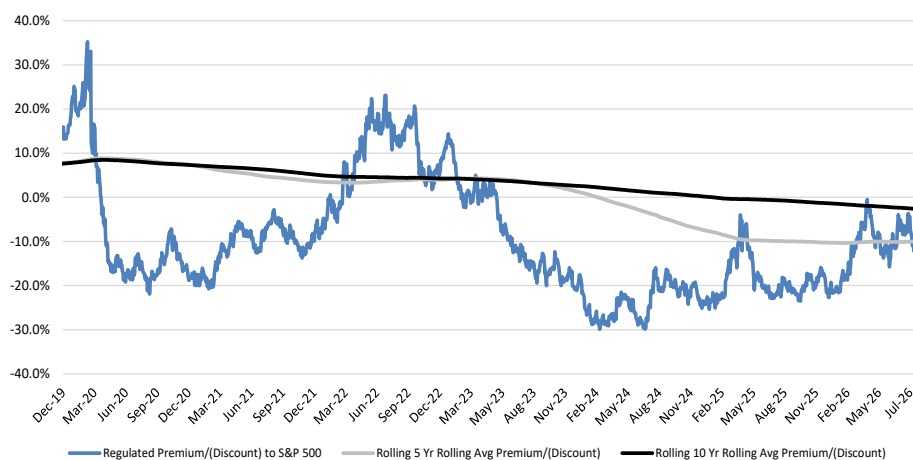
1) Where We Stand

Positive Outlook Given Surging Load Growth & Discounted Valuation

2026 Sees Ongoing Focus on Power Demand + Data Centers

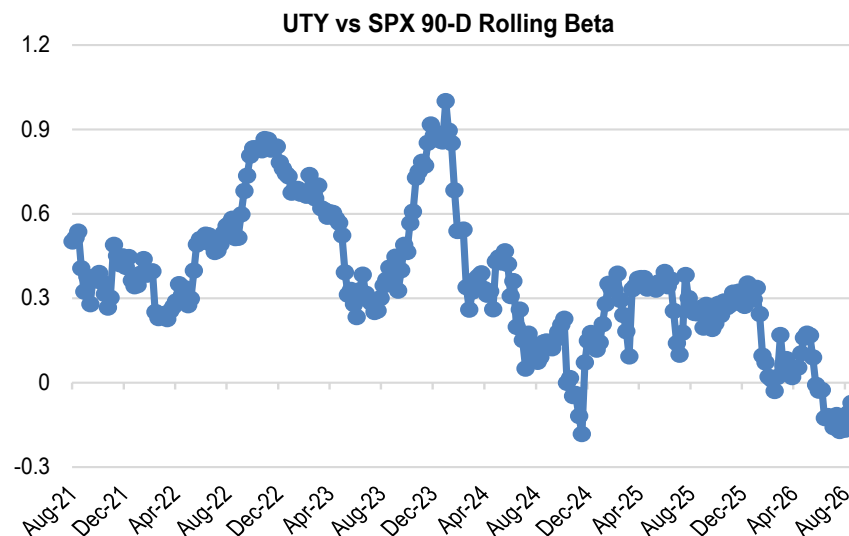
- LT tailwinds: demand growth from data centers, oil & gas/transportation/heating electrification, and manufacturing reshoring
- Low beta, defensive opportunity amid some macro uncertainty
- ~17.2x NTM regulated utilities P/E = -10% discount to the S&P 500
 - Historical -3% 10yr average discount
- Utilities offer steady compounding amid volatile market
 - Through 9/10, utilities (UTY) returned ~3% YTD (vs ~12% SPX)
- **Catalysts to watch through YE2026:** AEE, DTE, XEL, MDU
 - Upcoming IRPs for AEE and DTE to shape generation capex upside
 - 3Q capital plan refreshes for XEL and AEE in particular, with XEL citing \$10bn within line-of-sight and AEE citing 2.8GW in large load
 - Bakken East success could materially lift MDU's EPS CAGR

Utilities Stand at a Discount vs S&P 500



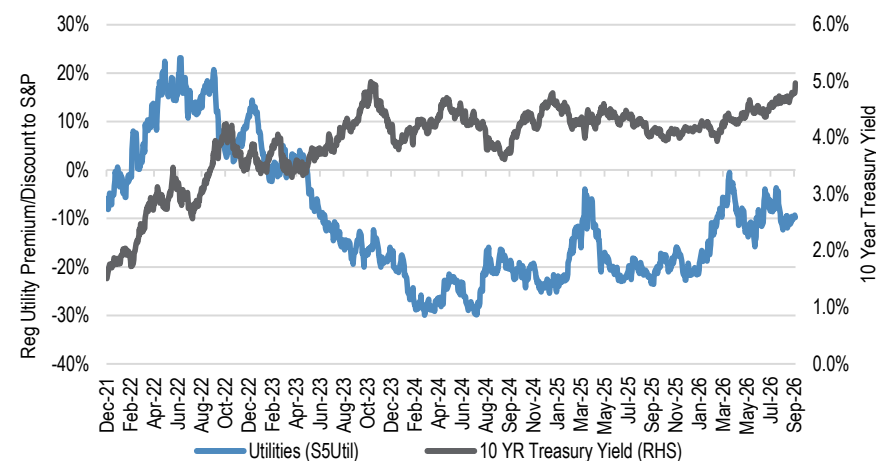
Source: Bloomberg Finance L.P.

UTY vs SPX 90-day Rolling Beta



Source: Bloomberg Finance L.P.

Utilities Performance (LHS) vs 10-year Yield (RHS)



Source: Bloomberg Finance L.P.

1) Where We Stand: Regression Reveals

Group Valuation Approaches Regression Implied Fair Value

Value Proposition Fair

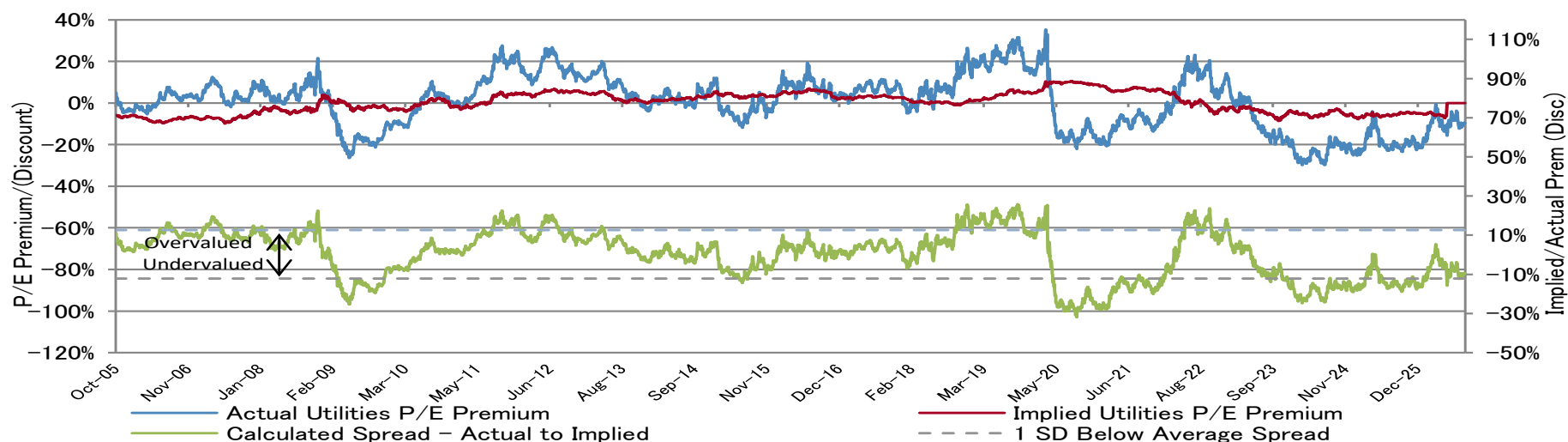
- Our 20-year regression versus the US 10-yr highlights group upside
 - Current analysis implies fair value at -8.5% discount (vs -9.7% actual)
 - See upside relative to recent ten-year average -3.0% discount
 - Ex-Mag 7 companies, utilities still stand at a -6% discount
 - 20 yr peak on 3/9/2020 at ~35% and peak discount of -30% on 7/8/2024
- Sector's discount screens attractive given strong fundamentals
 - A moderating rate environment could catalyze upside potential
 - Relative safe haven amidst any macroeconomic uncertainty
- JPM Global Equity Strategy team favors overweight utilities amid the volatile geopolitical environment.

JPM 20 yr Regression Implied Utility Relative Value

Current 10 Year Treasury Yield	4.96%
Current Relative Forward Premium	-9.73%
Time Length of Analysis	20 Years
Regression Equations	
Regulated/S&P500 Forward P/E =	$= -0.0429 (10 \text{ Year Yield}) + 0.1279$
Regression Input	Output: Implied Utility/SPX Forward Premium
Current 10 Year Treasury Yield	-8.5%
JPM Q3 2026 10 Year Treasury Yield Forecast (4.80%)	-7.8%
JPM Q4 2026 10 Year Treasury Yield Forecast (4.85%)	-8.0%

Source: Bloomberg Finance L.P.; Forecasted Premiums based on JPM Yield Forecasts Priced 9/10/2026; Utility Peer Group = 20 regulated electric companies

Spread Between Actual and 20 yr Regression Implied Utilities P/E Premium/Discount



Source: Bloomberg Finance L.P. Forecasted Premiums based on JPM Yield Forecasts in above table Priced 9/10/2026; Utility Peer Group = 20 regulated electric companies.

2) Favorite Names

ETR, AEE, XEL, PPL, NI, SRE

Entergy Corporation: (ETR): OW, PT: \$137

- Powerful June Investor Day update following a series of strong updates
 - ~13% EPS CAGR through 2030, and >8% expectations through 2035 off 2030 where we see levers to continue LDD growth
 - Boasts \$67bn capital plan, with high degree of [generation torque](#)
 - 16% industrial sales CAGR outlook, up from 15% at 4Q25
- Data centers: AWS in MS, for [Meta](#) in LA, and GOOG [in AR](#)
 - See [Meta](#) and [AWS](#) expansions proving out our clustering thesis
 - Continue to monitor expansion opportunities
- MS and LA [NARUC](#) conversations this summer reiterated stakeholder alignment for economic development via large load additions
- USGC industrial reshoring and resiliency capex tailwinds

Xcel Energy (XEL): OW, PT: \$97

- \$60bn plan [update](#) supports 6-8%+ EPS CAGR (9% over the plan period) and
 - ~11% rate base CAGR through 2030, which we view as sustainable with incremental capex opportunities
 - Recently outlined line of sight to \$10bn+ in upside opportunities for a 765kv line in SPP, 2,600MW in SPS RFP, 800MW in CO NTP, and Google in MN
 - See upside opportunity across remaining RFPs
- Recent rate case decisions reduce rate case exposure, providing a cleaner regulatory backdrop to refocus investor attention on growth
- While possessing data center growth, we see diversified growth drivers enabling a non-data center play
- CO and NM screened favorably in our recent [bill pressure analysis](#)

Ameren Corporation (AEE): OW, PT: \$133

- Expect noteworthy 3Q plan refresh following fall 2026 IRP filing
 - MO 2.8GW ESA signings represent substantive sales and EPS upside opportunity and capital plan investment visibility beyond 2030
 - [Transmission](#) project wins accentuate growth potential
- Messaging to not constrain growth to 6-8% EPS CAGR foreshadows strong EPS growth from conservative management team
- Moody's CFO-WC/Debt threshold reduction provides incremental financing flexibility
- Constructive MO [regulatory](#) environment and no MO gubernatorial election preferable to peers

PPL Corporation: (PPL): OW, PT: \$44

- Strong, 6-8% EPS growth with expectations to achieve high end
- Regulated generation capital investment opportunities screen comfortably against our analysis for [capex](#) increase required to maintain the company's ~10% rate base CAGR
- An Invitium JV deal announcement would serve as a catalyst
 - Company expects at least one deal announcement this year
 - View any success as advancing the scope and scale of PPL's long-term growth
 - Believe PJM uncertainty strengthens PPL's offering
- Regulatory exposure derisked following PA, KY, RI rate cases this year

3) Thoughts on Valuing TSR via PEGY [P/E / (G+Y)] Analysis

Utilizing PE/(G+Y) to Screen for Opportunities

ETR, XEL stand out among cleaner stories; XEL attractive despite wildfire concerns; MDU predicated on Bakken East

- We examine how much the market will pay (P/E) for expected total shareholder return (Y+G). P/E vs TR (PEGY)
- **XEL** attractive with cleaner regulatory backdrop than earlier this year
 - XEL exhibits a 1.27x PEGY ratio, compared to the 1.36x group median
 - ~9% EPS growth and near median dividend yield screens favorably in our analysis
 - Rate case orders territories derisk >70% of XEL's rate base, providing a cleaner regulatory backdrop against which we believe investor attention can refocus towards the company's 9% EPS CAGR, double digit rate base CAGR, and capital plan upside opportunities
- **ETR** offers attractive PEGY with lower regulatory/political risk
 - ETR's 1.36x PEGY (below peer median) ranks as one of the top electrics when removing riskier stories (i.e., major regulatory, CA)
 - 2030E/2028E EPS growth (11.9% vs 8.0%) more than makes up for a relatively lower dividend yield (2.5% vs 3.4%) compared to the peer set
- **MDU** exhibits a stand-out PEGY ratio when factoring in Bakken East EPS, though construction timing and financing terms remain key swing factors

Cali, ES, FE stand out as notable discount stocks; Canadians a mixed bag

- CA IOUs (**PCG**, **EIX**) exhibit lowest P/E to total return ratios, grounded by valuation discounts tied to notable wildfire risk and lack of durable policy reform
 - View **PCG**'s low valuation as compensating investors for risk premium and acknowledge substantive reform out of a special session could materially re-rate the stock
 - Continue to favor **PCG** over **EIX** given no exposure to the ongoing Eaton Fire investigation risks
- FE screens toward low end, but we see meaningful regulatory risk amid a period of elevated [region bill pressure](#)
- Canadians exhibit a wide range of valuations in our analysis
 - While higher yield vs US regulated peers provides support, we see limited earnings growth upside for the group from here
 - **FTS**'s P/E premium screens rich relative to US regulated names, given system investment upside more likely to hit in the 2030s
 - Despite **AQN** looking cheap at its current discount, we see opportunity for the company's outlook to normalize longer term.
 - **EMA**'s FL growth remains attractive, but we continue to see the balance sheet and high dividend payout as considerations within LT earnings growth
- **Methodology:** We calculate P/E to total shareholder return as (P/E) / (Y+G). On a closer look, we divide 2028 forward P/E by the sum of a two-year EPS CAGR from 2028-2030 and NTM dividend yield. Our framework utilizes a 2028 base year from which we calculate forward P/E and a two-year EPS CAGR to 2030 using JPM estimates. For the Canadians, we utilize 2028-2030 Bloomberg consensus.

3) Thoughts on Valuing TSR via PEGY [P/E /(G+Y)] Analysis

Utilizing PE/(G+Y) to Screen for Opportunities

PE/G and PETR Ratios Across Our Coverage

Company	Ticker			EPS Forecast			P/E Multiple	EPS Growth	PE/G	Yield	Yield + Growth	PETR	
		Rating	Price	2028E	2029E	2030E	2028E	2030/2028		NTM			
Regulated													
Ameren Corp	AEE	OW	\$106.41	6.39	6.96	7.59	16.7x	9.0%	1.9x	2.9%	11.9%	1.40x	
American Electric Power Co Inc	AEP	N	\$124.71	7.53	8.25	9.04	16.6x	9.6%	1.7x	3.1%	12.7%	1.30x	
CenterPoint Energy Inc	CNP	N	\$39.98	2.26	2.47	2.69	17.7x	9.0%	2.0x	2.4%	11.4%	1.56x	
CMS Energy Corp	CMS	OW	\$68.59	4.49	4.84	5.22	15.3x	7.8%	1.9x	3.4%	11.3%	1.36x	
DTE Energy Co	DTE	N	\$137.07	9.02	9.77	10.50	15.2x	7.9%	1.9x	3.5%	11.4%	1.33x	
Duke Energy Corp	DUK	N	\$121.40	7.63	8.17	8.73	15.9x	6.9%	2.3x	3.7%	10.6%	1.50x	
Consolidated Edison Inc	ED	UW	\$108.75	6.89	7.30	7.75	15.8x	6.1%	2.6x	3.3%	9.4%	1.68x	
Edison International	EIX	N	\$56.30	6.94	7.40	7.88	8.1x	6.5%	1.2x	6.4%	13.0%	0.63x	
Eversource Energy	ES	UW	\$71.47	5.22	5.56	5.91	13.7x	6.4%	2.1x	4.5%	10.9%	1.25x	
Entergy Corp	ETR	OW	\$107.78	5.85	6.55	7.33	18.4x	11.9%	1.5x	2.5%	14.4%	1.28x	
Exelon Corp	EXC	N	\$44.53	3.25	3.43	3.61	13.7x	5.4%	2.5x	3.9%	9.3%	1.48x	
FirstEnergy Corp	FE	N	\$47.04	3.19	3.45	3.74	14.7x	8.2%	1.8x	4.1%	12.3%	1.20x	
NiSource Inc	NI	OW	\$41.66	2.49	2.74	2.96	16.7x	9.1%	1.8x	3.0%	12.1%	1.38x	
PG&E Corp	PCG	OW	\$13.96	1.96	2.12	2.33	7.1x	9.0%	0.8x	1.8%	10.8%	0.66x	
PPL Corp	PPL	OW	\$35.11	2.29	2.47	2.67	15.3x	8.0%	1.9x	3.3%	11.3%	1.36x	
Public Service Enterprise Group Inc	PEG	N	\$73.81	5.05	5.44	5.73	14.6x	6.5%	2.2x	3.7%	10.2%	1.43x	
Southern Co/The	SO	N	\$88.77	5.36	5.78	6.22	16.6x	7.8%	2.1x	3.5%	11.2%	1.48x	
WEC Energy Group Inc	WEC	N	\$106.70	6.48	7.00	7.57	16.5x	8.0%	2.0x	3.7%	11.7%	1.40x	
Xcel Energy Inc	XEL	OW	\$76.34	5.02	5.47	5.95	15.2x	8.9%	1.7x	3.1%	12.0%	1.27x	
Mean							14.9x	8.0%	1.9x	3.5%	11.5%	1.31x	
Median							15.3x	8.0%	1.9x	3.4%	11.4%	1.36x	
Mean (ex EIX & PCG)							15.8x	8.0%	2.0x	3.4%	11.4%	1.39x	
Median (ex EIX & PCG)							15.8x	8.0%	1.9x	3.4%	11.4%	1.38x	
SMID													
MDU Resources Group, Inc.	MDU	OW	\$19.73	1.16	1.32	1.59	17.0x	16.8%	1.0x	3.1%	19.9%	0.85x	
Northwestern Energy Group Inc	NWE	N	\$70.90	4.23	4.41	4.59	16.8x	4.2%	4.0x	3.8%	8.0%	2.10x	
OGE Energy Corp	OGE	OW	\$46.89	2.79	2.98	3.19	16.8x	6.9%	2.4x	3.7%	10.6%	1.59x	
Portland General Electric Co	POR	N	\$49.17	3.76	3.93	4.32	13.1x	7.2%	1.8x	4.5%	11.7%	1.11x	
Mean							15.5x	6.1%	2.7x	4.0%	10.1%	1.60x	
Median							16.8x	6.9%	2.4x	3.8%	10.6%	1.59x	
Diversified													
Sempra	SRE	OW	\$84.57	6.00	6.53	7.21	14.1x	9.6%	1.5x	3.1%	12.8%	1.10x	
Mean (Regulated & Diversified)							14.9x	8.1%	1.9x	3.4%	11.5%	1.30x	
Median (Regulated & Diversified)							15.3x	8.0%	1.9x	3.4%	11.4%	1.36x	
Mean (Regulated & Diversified ex EIX/PCG)							15.7x	8.1%	2.0x	3.4%	11.5%	1.37x	
Median (Regulated & Diversified ex EIX/PCG)							15.6x	8.0%	1.9x	3.4%	11.4%	1.37x	
LDCs													
Atmos Energy Corp	ATO	OW	\$169.12	9.75	10.55	11.35	17.3x	7.9%	2.2x	2.5%	10.3%	1.68x	
New Jersey Resources Corp	NJR	OW	\$53.72	3.59	4.10	4.33	15.0x	9.8%	1.5x	3.7%	13.6%	1.10x	
Southwest Gas Holdings Inc	SWX	OW	\$88.69	5.68	6.70	7.25	15.6x	13.0%	1.2x	3.0%	15.9%	0.98x	
Spire Inc	SR	N	\$83.33	6.16	6.55	6.98	13.5x	6.4%	2.1x	4.1%	10.5%	1.29x	
Mean							16.0x	10.2%	1.6x	3.1%	13.3%	1.25x	
Median							15.6x	9.8%	1.5x	3.0%	13.6%	1.10x	
Canadians													
Algonquin Power & Utilities Corp	AQN	N	\$5.60	0.44	0.46	0.49	12.7x	5.3%	2.4x	4.7%	10.0%	1.27x	
Emera Inc	EMA	N	\$69.31	3.92	4.22	4.53	17.7x	7.5%	2.4x	4.3%	11.7%	1.51x	
Fortis Inc/Canada	FTS	UW	\$76.57	4.12	4.40	4.64	18.6x	6.2%	3.0x	3.5%	9.6%	1.93x	
Mean							16.3x	6.3%	2.6x	4.1%	10.5%	1.57x	
Median							17.7x	6.2%	2.4x	4.3%	10.0%	1.51x	
Water													
American Water Works Co Inc	AWK	N	\$140.89	7.09	7.72	8.23	19.9x	7.8%	2.6x	2.6%	10.3%	1.92x	
Mean							19.9x	7.8%	2.5x	2.6%	10.3%	1.92x	
Median							19.9x	7.8%	2.5x	2.6%	10.3%	1.92x	

Ticker	PETR
EIX	0.63x
PCG	0.66x
MDU	0.85x
SWX	0.98x
SRE	1.10x
NJR	1.10x
POR	1.11x
FE	1.20x
ES	1.25x
XEL	1.27x
AQN	1.27x
ETR	1.28x
SR	1.29x
AEP	1.30x
DTE	1.33x
CMS	1.36x
PPL	1.36x
NI	1.38x
AEE	1.40x
WEC	1.40x
PEG	1.43x
SO	1.48x
EXC	1.48x
DUK	1.50x
EMA	1.51x
CNP	1.56x
OGE	1.59x
ATO	1.68x
ED	1.68x
AWK	1.92x
FTS	1.93x
NWE	2.10x

Source: Bloomberg L.P., J.P. Morgan Estimates, Company Disclosures. Prices as of 9/3/2026. Please see our accompanying note [here](#)

4) Rate Base vs EPS Growth

Comparing Rate Base vs EPS Growth Targets after Equity Dilution

Degree of Difficulty in Hitting Targeted EPS Growth

- **ETR's** EPS growth rate cushion stands out due to ever-growing sales and generation investment upside
 - June Investor Day update raised implied rate base CAGR to 16.1%, with equity issuance sitting below 10% of an updated \$67bn plan, and yet again raised total retail sales CAGR to 9%
 - Data center expansion key to monitor given potential for hyperscalers to cluster, benefitting ETR's established toeholds across jurisdictions
- **SRE** shines here as well, underpinned by the convergence of rising TX T&D investments and zero common equity needs
 - Batch Zero wins represent additional upside
- **ATO** remains a leading rate base grower, primarily supported by safety-driven gas pipe replacement opportunities
- We utilize company-stated EPS and rate base growth targets, which can vary by time horizon and base year and potentially distort relative comparisons

Implied Financing/Regulatory Outcome Headroom

Ticker	Rating	Price	Rate Base CAGR	Estimated Dilution Annual Headwind	Implied EPS Growth post-Share Dilution	vs Targeted EPS CAGR	vs Targeted Sales Growth
ETR	OW	\$105.73	16.1%	2.8%	13.3%	>8%	9.0%
SRE	OW	\$84.00	11.0%	0.0%	11.0%	7-9%	
ATO	N	\$164.81	14.4%	4.7%	9.8%	6-8%	
CNP	N	\$39.12	11.0%	1.5%	9.5%	7-9%	
WEC	N	\$105.35	11.7%	3.0%	8.7%	7-8%	6-8%
FE	N	\$46.25	10.0%	1.5%	8.5%	6-8%	2.2%
OGE	OW	\$46.59	9.0%	0.7%	8.3%	5-7%	~3.5%
XEL	OW	\$75.41	11.0%	2.8%	8.2%	6-8%+	5.0%
AEP	N	\$123.47	10.9%	2.7%	8.1%	7-9%	
AEE	OW	\$104.95	10.6%	2.6%	8.0%	6-8%	6.2%
NI	OW	\$41.42	10.0%	2.4%	7.6%	9-10%	
DUK	N	\$119.37	9.6%	2.1%	7.5%	5-7%	3-4%
PPL	OW	\$34.21	10.3%	2.8%	7.5%	6-8%	
SO	N	\$87.75	9.0%	1.6%	7.4%	7-8%	10.0%
CMS	OW	\$67.68	10.5%	3.3%	7.2%	6-8%	3-3.5%
ES	UW	\$68.99	8.3%	1.3%	7.0%	5-7%	
EIX	N	\$56.75	7.0%	0.0%	7.0%	5-7%	
POR	N	\$48.56	15.0%	8.2%	6.8%	5-7%	3.0%
PEG	N	\$72.48	6.8%	0.0%	6.8%	6-8%	
AWK	N	\$139.84	8.5%	1.8%	6.7%	7-9%	
DTE	N	\$134.34	8.0%	1.9%	6.1%	6-8%	4-5%
ED	UW	\$106.73	8.8%	2.8%	6.0%	6-7%	~1%
EXC	N	\$43.39	7.9%	1.9%	6.0%	5-7%	3.3%
MDU	OW	\$19.24	7.5%	1.6%	5.9%	6-8%	
EMA	N	\$68.48	7.5%	1.9%	5.6%	5-7%	
FTS	UW	\$75.39	7.0%	1.6%	5.4%	NA	
NWE	N	\$70.01	5.0%	0.7%	4.3%	4-6%	

4) Rate Base vs Capital Plan Rolls

Five-Year Capital Plan Roll-Forward Expected to Support Rate Base CAGRs

Where We See Upside Potential in Gen Capex

- Ahead of companies' annual 5-year capital plan refresh, we size just how much incremental capex peers will need to maintain respective rate base CAGRs. . .
- AEE, AEP, XEL**: primed for meaningful capex upside
 - We see a strong case for capex upside for **AEE**, with 2.8GW in large load coming between 2027-2029
 - AEP** cites \$10bn of incremental capex within sight through 2030; near-term catalyst given 3Q refresh
 - Estimated ~15% plan increase for **XEL** to maintain its CAGR, but line-of-sight into \$13bn capex is reassuring

Awaiting the Capital Refresh: Time to Chin the Bar

- CMS, ATO, WEC** among higher implied capex hurdles
 - CMS**'s 10.5% rate base CAGR would demand a ~\$4-6bn capital plan increase; 2026 IRP may not command a shift change, but see other levers supporting EPS growth
 - ATO** needs to raise its capital plan by ~\$4bn to maintain rate base growth, approximately double the ~\$2bn ATO has increased its plan by in four of the last five refreshes
 - WEC**'s recent Point Beach PPA removes notable opportunity for additional owned generation capex
- Easy bar to chin, with some considerations: **POR, EXC, EIX**
 - Modest pullback in **POR**'s plan would sustain the CAGR, but WA deal and active GRC may be limiting factors
 - Flexible capital plan to achieve **EIX**'s ~7% rate base CAGR, though PCG's \$2bn capex cut sets a precedent
 - EXC** has a low bar to chin and line-of-sight into \$12-17bn of capex, but elevated regulatory/political risk in high bill pressure states may be a constraint

Companies Must Increase their Plan by ~8-10% to Maintain Growth

To maintain Rate Base CAGR:				Must Raise Capex Plan to:		
Ticker	Current Rate Base CAGR	Current 5-YR Capex Plan	Period	Required Roll-Forward Plan	Incremental \$ Required Δ	Implied Increase %
AEE	10.6%	\$32	2026 - 2030	\$35 - \$36	\$4	11.4%
AEP	~11%	\$78	2026 - 2030	\$77 - \$81	\$1	0.8%
ATO	13-15%	\$26	2026 - 2030	\$29 - \$31	\$4	15.5%
CMS	10.5%	\$24	2026 - 2030	\$28 - \$30	\$5	21.3%
CNP	11.0%	\$34	2026 - 2030	\$36 - \$38	\$3	9.4%
DUK	9.6%	\$103	2026 - 2030	\$110 - \$115	\$9	9.1%
ED	8.8%	\$38	2026 - 2030	\$39 - \$41	\$2	6.5%
EIX	~7%	\$40	2026 - 2030	\$37 - \$39	(\$1)	(2.8%)
EMA	7-8%	\$20	2026 - 2030	\$22 - \$23	\$2	11.8%
EXC	7.9%	\$42	2026 - 2029	\$39 - \$41	(\$2)	(4.0%)
FE	10.0%	\$36	2026 - 2030	\$38 - \$40	\$3	7.8%
FTS	7.0%	\$29	2026 - 2030	\$29 - \$31	\$1	3.3%
MDU	7-8%	\$3	2026 - 2030	\$3 - \$3	\$0	8.1%
OGE	9.0%	\$7	2026 - 2030	\$8 - \$8	\$1	10.3%
PEG	6-7.5%	\$24	2026 - 2030	\$25 - \$26	\$2	7.1%
POR	11.8%	\$8	2026 - 2030	\$7 - \$7	(\$0)	(6.5%)
PPL	10.3%	\$23	2026 - 2029	\$24 - \$26	\$2	9.0%
SO	9.0%	\$81	2026 - 2030	\$87 - \$92	\$8	10.4%
SR	7.0%	\$5	2026 - 2030	\$5 - \$5	\$0	6.9%
SRE	11.0%	\$65	2026 - 2030	\$70 - \$73	\$7	10.2%
SWX	9.5-11.5%	\$6	2026 - 2030	\$6 - \$7	\$0	1.4%
WEC	11.7%	\$38	2026 - 2030	\$41 - \$43	\$5	12.9%
XEL	~11%	\$60	2026 - 2030	\$67 - \$71	\$9	14.7%
Average						7.6%
Median						9.0%

Source: J.P. Morgan estimates. Please see our accompanying note [here](#)

5) VIU Generation Torque Analysis

Measuring Relative Degree of Opportunity Set for Additional Resource Additions for Select VIUs

ETR, WEC, and DUK Continue to Stand Out as Top Three

- **Methodology:** We take disclosed generation-specific capex under each company's current capital plan and divide by current market cap to showcase exposure to the power demand backdrop.
- **ETR** holds the highest generation capex exposure (~88% of market cap, and 67% of total capex), landing materially ahead of peers while simultaneously requiring modest equity needs at ~10% of its \$67bn capital plan
 - Continue to see data center expansion upon existing toeholds in ETR's service territories, as increasing focus on the delta between data center pipeline and equipment secured bodes well for ETR considering its 7.5GW of secured equipment
- **WEC** lands second, with generation capex representing ~59% of market cap (\$20.3bn 2026-30 generation capex reflecting ~54% of the capital plan)
 - While renewed Point Beach PPAs do not change the existing plan, we believe investors have come to expect a chunky increase in generation capex at WEC's 3Q plan update
- **DUK** tops the charts with its ~\$51bn total nominal generation spend, with the company's generation line-of-sight sitting at 15GW through 2031
- Looking for upcoming 3Q plan refresh cycle to bring generation capex upside into focus for **AEE** and **XEL**:
 - **AEE** points to a 60% increase in AEE MO sales by 2029 YE (vs. 2025) from 2.8GW in large load coming online, roughly doubling the current resource plan's 6.2% sales CAGR assumption, which we believe could translate to meaningful capex upside
 - We look ahead to the company's 2026 MO IRP, seeing the filing as a catalyst for AEE to refresh its generation needs and lift the capital plan
 - **XEL** messages \$10bn+ line-of-sight capital investment after adding \$6bn for 2,600MW company-owned resources in the SPS RFP
 - To the extent XEL can contract additional load beyond the 3GW targeted by YE26, we see additional investment opportunity where generation investment requirements land around \$2.5bn/GW

IRPs Present Significant Generation Upside for CMS & DTE; Additional Upside Considerations for the GenCo duo: PPL & NI

- Fall IRPs shape investment opportunity at **CMS** and **DTE**:
 - View upside relatively unchanged within the plan period, compared to 2Q messaging; however the IRP crystallizes visibility for CMS's generation opportunity set over the long-term
 - DTE points to ~\$5bn of incremental generation investments through 2032 before contemplating additional wins from ~2GW of hyperscaler load discussions and 3-4GW in other opportunities
- **PPL**: Invitium JV execution progresses, with generation sites, turbines, and interconnection queues representing \$6.4-7.7bn in capex after accounting for PPL's 51% share
 - We see PJM's dysfunction strengthening PPL's offering, supporting the company's expectations for one or multiple contract signings this year
- On the heels of GenCo, **NI** also appears poised for incremental wins, with strategic negotiations narrowing to 3GW (developing opportunities now at 2GW)

5) VIU Generation Torque Analysis

Measuring Relative Degree of Opportunity Set for Incremental Resource Additions

Generation Capex Ranker for Select Utilities						
Ticker	Gen Capex % of Mkt Cap	Gen Capex in Current Plan (\$bn)	Current Mkt Cap (\$bn)	Current Capex Plan Period	Load CAGR Guide	Potential Generation Capex Upside
ETR	88%	\$44.0	\$49.9	2026-2030	9.0%	Additional large customer load, including data centers and USGC industrial expansion, could drive incremental generation needs. Entergy's data center pipeline remains 7–12 GW, while other industries stand at 3-5GW. Not ready to update 7–12 GW figure, but cited customers' interest in expanding during the 2Q call
WEC	59%	\$20.3	\$34.6	2026-2030	6.0-8.0% (2028-2030)	Vantage's Port Washington campus (up to 3.5GW long term) and MSFT data center expansion underpin large power opportunities incremental to the current plan. Management noted 4-5GW ballpark for potential scale of both MSFT and Vantage projects, where additional generation to serve this load could cost \$2-2.5bn/GW.
DUK	54%	\$50.8	\$93.9	2026-2030	3-4% (2027-2029)	Solid load growth outlook w/ 7.8GW data center ESAs (+0.2GW from 1Q), and larger 15.4GW pipeline (including ESAs) expecting announcements over the course of 2026; new contracts signed in 2026 would ramp around the end of the plan period and could amount to \$5-10bn of additional capital above the existing plan
DTE	53%	\$15.0	\$28.3	2026-2030	4-5%	Identified ~\$5bn of incremental generation capex through 2032 related to servicing GOOGLE's 1GW project; see additional 2-3GW of hyperscaler load plus 3-4GW for multiple other opportunities; expect generation requirements to be solidified through IRP process
AEE	49%	\$14.5	\$29.4	2026-2030	6.2% (2026-2030)	2.8GW in signed ESAs with MO large load customers, representing upside to the 6.2% 2026-2030 sales CAGR; additional 0.6GW MO represents further sales growth and capital investment upside opportunities; executed ESAs have expressed interest in expanding their footprint; expect fall 2026 IRP filing and 3Q26 plan refresh
XEL	49%	\$23.5	\$47.4	2026-2030	5.0%	Achieved full line of sight into \$10bn+ incremental capital investment opportunity (up from \$7bn in 1Q), including 765kV in SPS, 800MW approved in PSCo, and ~3GW incremental data center demand; Full data center pipeline of +26GW (2GW currently under construction and expect to secure an additional 4GW by YE 2027); every 1GW of data center amounts to \$5-6bn of investments on generation side (incl 3rd party generation)
NI	45%	\$8.9	\$19.6	2026-2030		\$2Bn identified upside not included in Base Plan / GenCo; 9 GW GenCo data center opportunity, with signed capacity at ~4 GW, 3 GW in negotiation and 2 GW of developing opportunities. See opportunity and interest beyond 9 GW
CMS	41%	\$8.8	\$21.3	2026-2030	2-3%	Cited \$3bn upside not included in the current plan; Data centers and the 2026 IRP represent incremental capital opportunities; each 1GW additional load onto the system represents \$2-5bn capital investment
AEP	36%	\$24.0	\$66.9	2026-2030		Telegraphed a line of sight to at least \$10bn in additional opportunities within 2026-2030 (Piketon transmission and Wyoming fuel cell together representing ~\$8bn, plus incremental generation); new generation capex could emerge should large load demand drive resource needs in AEP's vertically integrated states
POR	33%	\$1.9	\$5.8	2026-2030	3% through 2030	Capex forecast includes 2023 RFP; 2025 RFP and 1.7 GW incremental large load represents upside to plan, with projects expected to COD by YE30; WA Utility acquisition would add generation assets
SO	29%	\$28.4	\$99.4	2026-2030	~10% average annual growth (2026-2030)	Quantified ~\$2.5bn/GW incremental generation; GA RFP filed with initial 2-6GW scale. 17GW contracted large load pipeline, up from 11GW in 1Q, progressed an additional 11GW to next stage; 700MW uprate opportunities identified during the 4Q25 call
PPL	22%	\$5.7	\$25.9	2026-2029	>3.0% (KY 2024-2032 mid-load scenario)	Incremental \$10-12bn upside from additional KY generation needs and the Blackstone JV through 2032; 31.8GW signed agreement pipeline (up from 28.3GW); KY: load pipeline of 13.7GW by 2032 (up from 12.9GW); Higher demand indicates a CPCN for additional gen resources will be filed by YE, reflecting \$3.5-4bn of incremental investment need (2027 -2032)
NWE	17%	\$0.7	\$4.3	2026-2030		Incremental opportunities in data centers/large loads, FERC regional transmission, and generating capacity; pipeline features three development agreements, yet to sign ESA with large load customers
OGE	16%	\$1.5	\$9.6	2026-2030		Look for 1) RFP bid selections to support OG&E's identified capacity needs, with a focus on the size of utility-owned capex opportunities, and 2) final NTCs for the SPP ITP Seminole–Shreveport 765-kV line; we estimate a hypothetical \$1.4-2.4bn total RFP capex opportunity through 2032 should OG&E satisfy ~50% of its capacity need with utility-owned gas generation

Source: Company filings; we assume 100% of SO Power's 2026-2030 capex relates to new generation. Market cap as of 9/2/2026. Please see our accompanying note [here](#)

6) Bill Pressure and O&M Analysis: Key Lever to Manage Bill Headroom

A Look at Bill Directionality in 2026

PJM Pain Persists, but PPL Positioning Preferred

- U.S. combined average utility bills increased ~5% for 1H26/1H25 relative to last year's winter season (~6% electric, ~3% gas YoY)
- **AK, DE, HI, NY, and NH** lead the pack for highest rate of change in total utility bills in the first six months of 2026
 - Continue to see more downside than upside in **NJ**; market does not appear to fully appreciate the range of outcomes policymakers may pursue to unlock ratepayer savings
 - **NJ**: expect affordability overhang for **PEG, EXC** and **FE**
 - **NY's** double-digit bill increases pose headline risk to **ED** and breach Gov's goal to keep bills at pace of inflation
- Multiple **PJM** states land in the high-end of our screen, illustrating continued pressure in the region (**MD, DC, PA, NJ, VA, and OH**)
 - **PA**: see **PPL** positioned relatively favorably following recent rate case order
- Expect affordability to remain extremely topical, as multiple states gear up for gubernatorial elections later this year

States with Less Risk, in Our View

- **NM, NV, CO, WY, and OR** experienced the greatest recent bill declines in our analysis
 - **NV's** and **AZ's** favorable trajectory screens positively for **SWX**, with dual GRC filings in NV & AZ
- **CO** and **NM** on the lower end of our analysis screens positive for **XEL**
- **OR** electric bills decline stands as a positive signal of progress behind **POR's** ability to manage rate levels ahead of its next GRC
 - Nevertheless, we acknowledge this trend follows a period of high electric rate of change in 2024

YoY Change in Average Monthly Utility Bills per Customer

State	1H2026 YoY		
	Combined	Electric	Gas
Alaska	26.6%	13.4%	39.7%
Delaware	16.6%	8.2%	30.0%
Hawaii	16.4%	12.5%	25.5%
New York	15.3%	15.1%	15.6%
New Hampshire	15.0%	17.0%	12.8%
Maine	13.6%	10.7%	16.6%
Maryland	13.4%	25.3%	-4.8%
District of Columbia	12.9%	18.0%	7.6%
Pennsylvania	11.7%	13.7%	9.1%
New Jersey	11.5%	15.2%	7.8%
Virginia	11.3%	11.5%	11.1%
Ohio	11.2%	10.6%	12.0%
Kentucky	8.7%	6.5%	12.6%
Wisconsin	7.6%	3.7%	13.2%
Vermont	7.4%	6.8%	8.0%
North Dakota	6.5%	6.4%	6.7%
South Carolina	6.4%	4.5%	10.2%
Massachusetts	6.3%	0.5%	11.6%
Michigan	6.3%	6.2%	6.4%
Iowa	5.7%	3.0%	10.1%
South Dakota	5.1%	5.9%	3.3%
North Carolina	4.9%	6.0%	2.9%
Tennessee	4.5%	2.9%	8.1%
Mississippi	4.2%	5.1%	2.4%
Indiana	4.1%	4.9%	2.7%
Nebraska	4.1%	4.0%	4.3%
Illinois	3.9%	5.3%	2.4%
Oklahoma	3.6%	2.8%	5.0%
Connecticut	3.4%	-6.3%	17.1%
Minnesota	3.0%	3.0%	2.9%
Washington	2.3%	8.8%	-5.3%
West Virginia	2.2%	-1.1%	7.7%
Arkansas	2.1%	2.5%	1.5%
Florida	1.6%	1.3%	2.7%
Georgia	1.4%	0.5%	2.5%
Kansas	1.1%	3.7%	-1.9%
Texas	0.0%	1.5%	-3.2%
Arizona	-0.1%	5.2%	-13.8%
Utah	-0.2%	1.0%	-2.0%
Missouri	-0.3%	3.0%	-4.5%
Idaho	-0.5%	3.9%	-10.2%
California	-0.8%	5.6%	-13.1%
Alabama	-0.9%	-0.8%	-1.2%
Rhode Island	-1.9%	-2.2%	-1.5%
Louisiana	-2.2%	2.6%	-13.9%
Montana	-3.5%	4.8%	-17.4%
Oregon	-4.1%	-3.2%	-5.4%
Wyoming	-4.7%	-0.2%	-10.7%
Colorado	-5.2%	4.5%	-18.5%
Nevada	-7.8%	4.7%	-30.2%
New Mexico	-10.4%	-1.6%	-23.9%
US Average	4.7%	5.6%	3.0%

Source: EIA, J.P. Morgan estimates. Please see our accompanying note **J.P.Morgan**

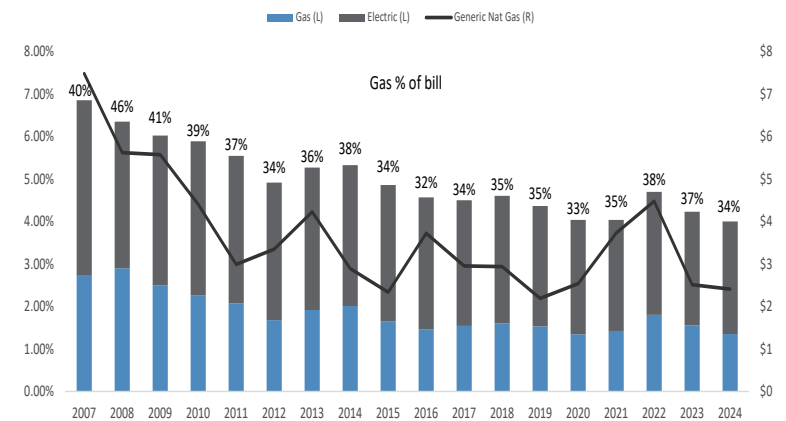
6) Bill Pressure and O&M Analysis: Key Lever to Manage Bill Headroom

Bill Pressure Ranker 2024 Trends

State-Level Dynamics Showcase a Wide Range of Outlooks

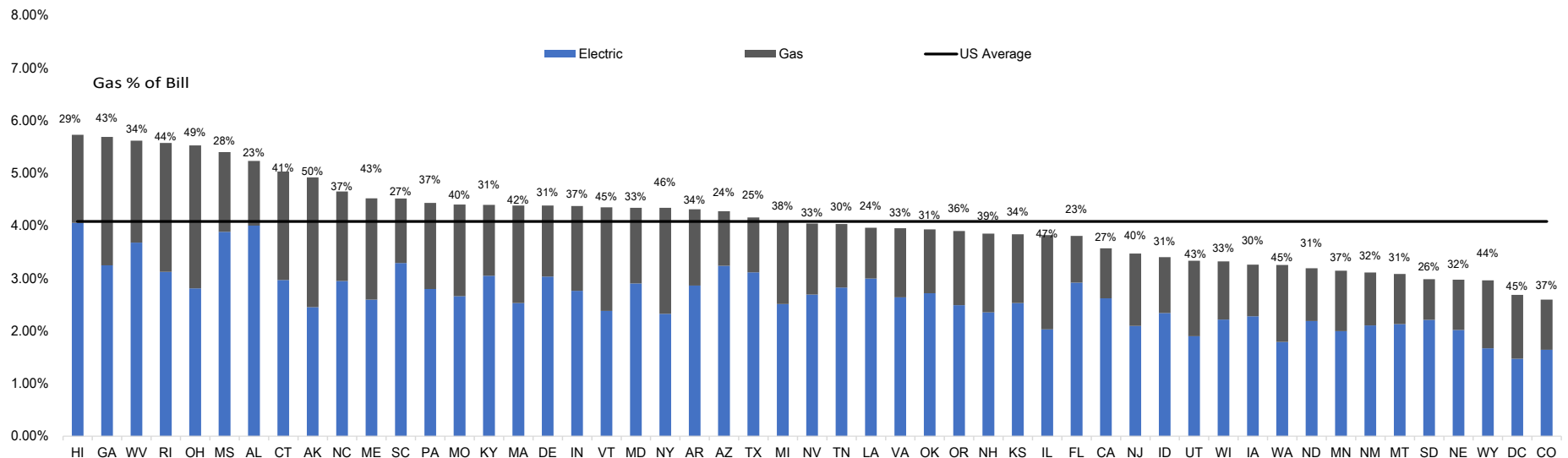
- 2024 wallet share: ~4.1% US avg vs ~4.3% in 2023
 - Indicates headroom versus peak levels of nearly 7% in 2008
 - As % per capita disposable income: **HI, GA, WV, RI, and OH** screen most stressed; **CO, DC, WY, NE, and SD** have the lowest wallet share
 - Lagged nature of the data does not yet capture higher YTD gas and power prices
 - Expect data center tariff and cost allocation agreements to remain under scrutiny
- Northeast in spotlight with energy supply challenges and PJM auction impacts
 - Within the Northeast, **RI, CT, and ME** carry the highest wallet share
 - **GA, MS, and AL** screen relatively weak on this metric, though we note favorable regulatory environments in these states as considerations here.
 - **OH** leads PJM as the highest wallet share state, primarily due to a high gas bill portion of combined bills (which carries greater volatility).

Historical Electric and Gas Utility Bill as a Percentage of Per Capita Disposable Income, U.S. Average 2007-2024



Source: J.P. Morgan

Electric & Gas Utility Bill as a % of Per Capita Disposable Income by State 2024



Source: EIA, BEA, J.P. Morgan estimates, J.P. Morgan Utilities Credit Research Team; Please see accompanying note [here](#)

J.P.Morgan

6) Bill Pressure and O&M Analysis: Key Lever to Manage Bill Headroom

State-Level Bill Pressure Metrics – Most Stressed 2024

- **Higher stress screen:** 1) below average per capita disposable income, 2) above average utility bills as a percentage of disposable income, 3) YoY increases in combined utility bills
- **AZ, KY, RI, SC, VT, and WV** stand out based on aforementioned screen and RRA filter that excludes in-line or better jurisdictions
 - **RI** takes the spotlight with a wallet share ~150bps above the national average and a +8.15% YoY increase in combined bills
 - **KY** stands out as a sensitive state given the low income profile (though PPL's specific service territory screens well within the state)
 - **WV** remains highly sensitive to bill pressures given its rather challenging regulatory environment (as demonstrated in AEP's last rate order)
- When including all RRA ranking, **GA, NC, and OH** land in our screen based on the aforementioned metrics
 - **GA** stands as the U.S. continental state leader for wallet share at ~5.7% and we track a degree of stakeholder pushback on authorized ROEs
 - **OH** exhibits a notably high wallet share size (~150bp above avg) and a notably high spike in combined utility bills (+28% YoY)

Select State-Level Bill Pressure Metrics – Most Stressed Forced Ranker (No RRA Filter)

State	2024 Per Capital Disposable Income as % of National Average	Combined Utility Bill as % of Per Capital Disposable Income	YoY Δ in Combined Utility Bills	YoY Δ in Electric Bills	YoY Δ in Gas Bills	Regulatory Environment (Commission Ranking)	Frequency of Fuel Adjustment
National Average	\$63,366	4.1%	0.3%	3.7%	-4.7%		
Georgia	88%	5.7%	8.9%	6.8%	12.0%	Above Average / 2	Undefined
Ohio	91%	5.5%	27.5%	8.4%	55.8%	Average / 1	1-2x / year
Rhode Island	98%	5.6%	8.2%	8.4%	7.8%	Average / 2	1-2x / year
West Virginia	80%	5.6%	6.5%	11.7%	-2.1%	Below Average / 1	1-2x / year
North Carolina	92%	4.7%	17.6%	12.3%	28.1%	Above Average / 3	1-2x / year
South Carolina	86%	4.5%	5.3%	6.9%	1.3%	Average / 2	2x+ / year

State-Level Bill Pressure Metrics – Most Stressed (RRA State Ranking Filter)

State	2024 Per Capital Disposable Income as % of National Average	Combined Utility Bill as % of Per Capital Disposable Income	YoY Δ in Combined Utility Bills	YoY Δ in Electric Bills	YoY Δ in Gas Bills	Regulatory Environment (Commission Ranking)	Frequency of Fuel Adjustment
National Average	\$63,366	4.1%	0.3%	3.7%	-4.7%		
Arizona	93.6%	4.3%	5.0%	7.9%	-3.4%	Below Average / 2	Undefined
Kentucky	83.1%	4.4%	0.3%	6.4%	-11.1%	Average / 2	2x+ / year
Rhode Island	98.5%	5.6%	8.2%	8.4%	7.8%	Average / 2	1-2x / year
South Carolina	86.0%	4.5%	5.3%	6.9%	1.3%	Average / 2	2x+ / year
Vermont	99.9%	4.4%	3.0%	7.3%	-1.7%	Average / 3	2x+ / year
West Virginia	79.6%	5.6%	6.5%	11.7%	-2.1%	Below Average / 1	1-2x / year

Source: EIA, BEA, S&P, J.P. Morgan estimates, J.P. Morgan Utilities Credit Research Team. Note that the color-coded heat map is relative to all states.

6) Bill Pressure and O&M Analysis: Key Lever to Manage Bill Headroom

State-Level Bill Pressure Metrics – Least Stressed 2024

- **Low stress screen:** 1) above average per capita disposable income, 2) below average utility bills as a percentage of disposable income, 3) YoY declines in combined utility bills by, 4) removed states with “Average/3” or lower commission rankings from consideration
- Midwest (**CO**, **MN**, **NE**, **ND**, and **SD**) screens as the most favorable region, featuring high-income and/or declining bill states
 - **MN**’s significant pullback in customer bills, potentially bodes well for XEL’s bill headroom and generally more positive regulatory relationship
 - **CO** and **WY**’s high income metrics should support long-term bill headroom net of the YoY electric bill increases
 - Our recent CO PUC discussion [voiced](#) a clear focus on prioritizing productive, value-creating investments that tangibly benefit customers
- **NJ** carries elevated affordability concerns despite the state’s relatively low wallet share given a +13% YoY spike in electric bills and pronounced governor-elect rhetoric on resolving bill pressures (extended time before another rate case).
- **MD** and **OR** face somewhat higher bill pressures in given each state’s double-digit YoY electric bill increases
- **CT** also remains a highly sensitive state given high wallet share and an uncertain regulatory climate following recent PURA changes and the Aquarion sale [rejection](#).

State-Level Bill Pressure Metrics – Least Stressed

State	2024 Per Capital Disposable Income as % of National Average	Combined Utility Bill as % of Per Capital Disposable Income	YoY Δ in Combined Utility Bills	YoY Δ in Electric Bills	YoY Δ in Gas Bills	Regulatory Environment (Commission Ranking)	Frequency of Fuel Adjustment
National Average	\$63,366	4.1%	0.3%	3.7%	-4.7%		
Colorado	116%	2.6%	-4.6%	6.3%	-18.9%	Average / 1	1-2x / year
Minnesota	104%	3.1%	-8.7%	-0.7%	-19.9%	Average / 2	2x+ / year
Nebraska	103%	3.0%	-13.1%	-1.1%	-30.9%	Average / 2	Undefined
North Dakota	102%	3.2%	-7.0%	0.6%	-20.1%	Average / 1	2x+ / year
South Dakota	110%	3.0%	-5.2%	1.2%	-19.7%	Average / 2	Undefined

Other Select States with Notable Bill Pressure

State	2024 Per Capital Disposable Income as % of National Average	Combined Utility Bill as % of Per Capital Disposable Income	YoY Δ in Combined Utility Bills	YoY Δ in Electric Bills	YoY Δ in Gas Bills	Regulatory Environment (Commission Ranking)	Frequency of Fuel Adjustment
National Average	\$63,366	4.1%	0.3%	3.7%	-4.7%		
Connecticut	127%	5.0%	3.3%	-1.5%	11.1%	Below Average / 3	2x+ / year
Maryland	108%	4.3%	5.8%	11.7%	-4.5%	Below Average / 3	1-2x / year
New Jersey	116%	3.5%	9.7%	13.2%	4.9%	Average / 3	1-2x / year
Oregon	99%	3.9%	4.6%	10.2%	-3.9%	Average / 3	1-2x / year

Source: EIA, BEA, S&P, J.P. Morgan estimates, J.P. Morgan Utilities Credit Research Team

6) Bill Pressure and O&M Analysis: Key Lever to Manage Bill Headroom

State-Level Bill Pressure Metrics and YoY Change in Wallet Share Utility Bills (2024 vs 2023)

State	2024 Per Capita disposable income	Electric + Gas Bill as % of per capita disposable income	Gas % of the Bill	Regulatory Environment (Commission Ranking)	Frequency of fuel adjustment	One-Year Delta					
						Change in Wallet Share (Combined)	Combined Annual Bill YoY	Electric Bill YoY	Gas Bill YoY	Median Income YoY	
National Avg	\$ 63,366	4.08%	35%								
Alabama	\$ 51,997	5.23%	23%	Above Average / 1	Undefined						
Alaska	\$ 70,193	4.92%	50%	Below Average / 1	1-2x / year						
Arizona	\$ 59,339	4.28%	24%	Below Average / 2	Undefined						
Arkansas	\$ 54,137	4.31%	34%	Average / 1	1-2x / year						
California	\$ 73,684	3.57%	27%	Average / 2	1-2x / year						
Colorado	\$ 73,513	2.59%	37%	Average / 1	1-2x / year						
Connecticut	\$ 80,694	5.03%	41%	Below Average / 3	2x+ / year						
Delaware	\$ 59,650	4.38%	31%	Average / 2	1-2x / year						
District of Columbia	\$ 92,365	2.69%	45%	Below Average / 1	1-2x / year						
Florida	\$ 64,167	3.81%	23%	Above Average / 2	2x+ / year						
Georgia	\$ 55,861	5.69%	43%	Above Average / 2	Undefined						
Hawaii	\$ 62,780	5.73%	29%	Average / 2	2x+ / year						
Idaho	\$ 55,707	3.40%	31%	Average / 2	Undefined						
Illinois	\$ 64,962	3.82%	47%	Average / 3	2x+ / year						
Indiana	\$ 57,736	4.38%	37%	Average / 1	2x+ / year						
Iowa	\$ 58,732	3.26%	30%	Above Average / 3	2x+ / year						
Kansas	\$ 58,737	3.84%	34%	Average / 3	2x+ / year						
Kentucky	\$ 52,673	4.40%	31%	Average / 2	2x+ / year						
Louisiana	\$ 56,469	3.96%	24%	Average / 3	2x+ / year						
Maine	\$ 61,744	4.52%	43%	Average / 3	Undefined						
Maryland	\$ 68,513	4.34%	33%	Below Average / 3	1-2x / year						
Massachusetts	\$ 79,289	4.39%	42%	Average / 2	1-2x / year						
Michigan	\$ 56,984	4.07%	38%	Average / 1	1-2x / year						
Minnesota	\$ 66,131	3.15%	37%	Average / 2	2x+ / year						
Mississippi	\$ 47,831	5.40%	28%	Above Average / 3	2x+ / year						
Missouri	\$ 58,274	4.41%	40%	Average / 2	1-2x / year						
Montana	\$ 60,758	3.08%	31%	Average / 3	2x+ / year						
Nebraska	\$ 65,534	2.97%	32%	Average / 2	Undefined						
Nevada	\$ 62,129	4.04%	33%	Average / 1	2x+ / year						
New Hampshire	\$ 73,890	3.85%	39%	Average / 3	Undefined						
New Jersey	\$ 73,392	3.47%	40%	Average / 3	1-2x / year						
New Mexico	\$ 52,944	3.11%	32%	Below Average / 1	2x+ / year						
New York	\$ 72,029	4.34%	46%	Average / 2	2x+ / year						
North Carolina	\$ 58,384	4.65%	37%	Above Average / 3	1-2x / year						
North Dakota	\$ 64,891	3.19%	31%	Average / 1	2x+ / year						
Ohio	\$ 57,734	5.53%	49%	Average / 1	1-2x / year						
Oklahoma	\$ 58,282	3.93%	31%	Average / 3	2x+ / year						
Oregon	\$ 62,482	3.90%	36%	Average / 3	1-2x / year						
Pennsylvania	\$ 62,291	4.43%	37%	Above Average / 2	1-2x / year						
Rhode Island	\$ 62,392	5.57%	44%	Average / 2	1-2x / year						
South Carolina	\$ 54,492	4.52%	27%	Average / 2	2x+ / year						
South Dakota	\$ 69,389	2.98%	26%	Average / 2	Undefined						
Tennessee	\$ 60,872	4.03%	30%	Above Average / 3	Undefined						
Texas	\$ 63,072	4.16%	25%	Below Average / 1	1-2x / year						
Utah	\$ 59,663	3.33%	43%	Average / 2	1-2x / year						
Vermont	\$ 63,291	4.35%	45%	Average / 3	2x+ / year						
Virginia	\$ 67,615	3.95%	33%	Average / 1	2x+ / year						
Washington	\$ 76,112	3.26%	45%	Average / 3	Undefined						
West Virginia	\$ 50,444	5.62%	34%	Below Average / 1	1-2x / year						
Wisconsin	\$ 60,040	3.32%	33%	Above Average / 3	2x+ / year						
Wyoming	\$ 77,361	2.96%	44%	Average / 2	Undefined						
Maine						-1.37%	-18.49%	-13.00%	-24.88%		6.25%
New Hampshire						-0.87%	-13.64%	-14.17%	-12.78%		5.83%
Montana						-0.56%	-10.01%	-1.45%	-24.66%		6.41%
Hawaii						-0.54%	-1.58%	-0.52%	-4.05%		7.72%
Nebraska						-0.52%	-13.15%	-1.12%	-30.91%		2.06%
Oklahoma						-0.48%	-6.44%	2.28%	-21.45%		5.08%
Nevada						-0.48%	-5.20%	-4.28%	-6.97%		6.09%
New Mexico						-0.47%	-8.02%	1.84%	-23.50%		5.95%
Minnesota						-0.46%	-8.67%	-0.65%	-19.92%		4.55%
Iowa						-0.43%	-8.22%	-0.95%	-21.58%		3.85%
Idaho						-0.38%	-5.41%	1.95%	-18.42%		5.16%
Florida						-0.36%	-3.77%	-7.29%	9.94%		5.24%
Arkansas						-0.35%	-4.30%	0.49%	-12.53%		3.55%
Utah						-0.31%	-2.60%	11.29%	-16.45%		6.47%
Wisconsin						-0.30%	-4.74%	-0.17%	-12.72%		3.82%
South Dakota						-0.30%	-5.23%	1.19%	-19.75%		4.18%
Texas						-0.26%	-1.34%	-1.26%	-1.56%		4.86%
Colorado						-0.25%	-4.59%	6.26%	-18.85%		4.75%
Kentucky						-0.24%	0.30%	6.36%	-11.14%		5.89%
Missouri						-0.24%	-1.25%	2.45%	-6.41%		4.18%
California						-0.23%	0.08%	11.08%	-21.36%		6.57%
North Dakota						-0.22%	-6.98%	0.58%	-20.09%		-0.53%
Pennsylvania						-0.21%	-2.16%	1.45%	-7.77%		2.43%
Louisiana						-0.20%	0.19%	-1.40%	5.46%		5.30%
Kansas						-0.19%	-5.28%	5.08%	-20.46%		-0.62%
Michigan						-0.18%	-0.09%	5.01%	-7.36%		4.34%
Virginia						-0.18%	0.29%	5.04%	-8.07%		4.82%
Alabama						-0.17%	2.93%	6.66%	-7.60%		6.24%
Massachusetts						-0.15%	0.64%	0.99%	0.17%		4.18%
Wyoming						-0.15%	1.34%	8.47%	-6.59%		6.41%
Vermont						-0.14%	3.04%	7.29%	-1.67%		6.45%
Connecticut						-0.13%	3.29%	-1.52%	11.12%		6.02%
Alaska						-0.11%	4.10%	4.10%	4.10%		6.44%
Mississippi						-0.09%	2.68%	1.15%	6.81%		4.42%
New York						-0.09%	3.11%	10.91%	-4.63%		5.25%
Delaware						-0.07%	-0.21%	9.29%	-16.54%		1.47%
Oregon						-0.07%	4.61%	10.17%	-3.94%		6.56%
South Carolina						-0.04%	5.30%	6.86%	1.34%		6.28%
Washington						-0.03%	5.73%	5.90%	5.53%		6.86%
Arizona						-0.02%	4.96%	7.94%	-3.40%		5.41%
Maryland						-0.01%	5.80%	11.73%	-4.46%		5.92%
District of Columbia						0.05%	5.04%	9.03%	0.58%		3.16%
Indiana						0.08%	7.55%	1.65%	19.44%		5.52%
West Virginia						0.08%	6.50%	11.69%	-2.13%		4.93%
Tennessee						0.13%	10.57%	5.98%	23.03%		6.91%
Illinois						0.15%	7.26%	4.65%	10.37%		3.08%
Georgia						0.17%	8.94%	6.76%	11.99%		5.64%
Rhode Island						0.20%	8.15%	8.43%	7.81%		4.30%
New Jersey						0.21%	9.74%	13.18%	4.90%		3.13%
North Carolina						0.45%	17.61%	12.29%	28.13%		6.16%
Ohio						0.97%	27.46%	8.40%	55.76%		5.13%
National Average						-0.18%	0.31%	3.69%	-4.67%		4.90%

Source: EIA, BEA, J.P. Morgan estimates.

J.P.Morgan

6) Bill Pressure and O&M Analysis: Key Lever to Manage Bill Headroom

O&M Represents an Important Component to Utility Flexibility

Measuring O&M Performance in Our Utility Coverage Universe

- With an uptick in investor focus on any potential changing regulatory tides given growing consumer affordability concerns amidst data center growth (as seen in IN), expense efficiency stands out as a paramount consideration within bill pressure discussions
- Load and other system investment needs continue to drive higher capex plans, where a backstop from demonstrated track records on O&M management remains a clear tool to de-risk plans and guard against rate-related downside regulatory outcomes
- Companies that can credibly point to historical expense management may garner more investor confidence in plan execution ability
- Our historical cost structure analysis utilizes FERC Form 1 data to analyze electric utility cost trends
- Updating the rankings, with historical rolling average O&M trends now favoring **CNP**
 - **CNP** stands out on both a three- and five-year basis given O&M reductions ~2% year over year since 2021 and the company's continued focus on reducing O&M 1-2% through 2030

Top Utilities by Category

Non-fuel O&M	Adjusted non-fuel O&M	SG&A	Adjusted SG&A
Top Over 3 years			
PCG	CNP	PCG	CNP
D	D	D	D
DTE	DTE	CNP	WEC
Top Over 5 years			
PCG	NEE	PCG	CNP
NEE	CNP	ED	WEC
DUK	DUK	WEC	PPL

Non-Fuel O&M per Customer

Ticker	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Rolling Avg CAGR
Largest Decline Over Three Years											
PCG	747	821	697	2,952	3,052	1,363	1,533	1,512	1,666	1,321	-8.9%
D	682	751	561	558	1,113	794	532	486	775	913	-3.8%
DTE	729	792	765	775	758	771	789	757	712	715	-2.0%
DUK	739	723	665	667	695	666	552	417	686	731	-1.4%
ED	701	681	632	633	635	581	610	623	554	613	-0.7%
Mean											1.9%
Median											1.7%
Largest Decline Over Five Years											
PCG	747	821	697	2,952	3,052	1,363	1,533	1,512	1,666	1,321	-7.7%
NEE	419	367	623	318	402	332	225	68	512	463	-4.9%
DUK	739	723	665	667	695	666	552	417	686	731	-2.0%
WEC	1,068	1,083	1,014	1,089	1,015	947	933	866	1,005	1,011	-1.6%
ED	701	681	632	633	635	581	610	623	554	613	-1.2%
Mean											0.9%
Median											0.8%

Source: FERC Form 1, S&P Capital IQ, J.P. Morgan adjustments. Please see our accompanying note [here](#).

6) Bill Pressure and O&M Analysis: Key Lever to Manage Bill Headroom

Metric Summary of Different Metrics Utilized in Our Cost Structure Analysis

- **Electric Non-Fuel O&M Calculation:** We start with total electric O&M and adjust out all fuel and purchased power costs. We then compare this output on a per retail customer basis.
- **Adjusted Electric Non-Fuel O&M Calculation:** We start with total electric O&M and adjust out all fuel and purchased power costs. Additional adjustments include employee pensions and benefits, injuries and damages, and transmission of electricity by others. We then compare this output on a per retail customer basis.
- **SG&A Calculation:** For our SG&A metric, we add 1) total customer accounts expenses, 2) total customer service and information expenses, 3) total sales expenses, and 4) total administrative and general expenses. We then compare this output on a per retail customer basis.
- **Adjusted SG&A Calculation:** For our adjusted SG&A metric, we add 1) total customer accounts expenses, 2) total customer service and information expenses, 3) total sales expenses, and 4) total administrative and general expenses. Subsequently, we remove employee pensions & benefits and injuries & damage expenses. We then compare this output on a per retail customer basis.

Adjusted Non-Fuel O&M \$ per Customer, 3- and 5-year Decreases

Ticker	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Rolling Avg CAGR
Largest Decline Over Three Years											
CNP	262	262	260	279	276	258	244	235	230	248	-2.9%
D	635	708	518	487	1,015	753	509	474	756	880	-2.5%
DTE	532	568	540	558	547	556	542	540	497	488	-2.5%
DUK	683	659	607	609	641	616	505	373	652	695	-0.8%
PPL	507	495	473	497	493	460	492	508	474	495	0.7%
Mean											3.1%
Median											3.4%
Largest Decline Over Five Years											
NEE	388	341	597	294	377	314	204	47	495	447	-4.8%
CNP	262	262	260	279	276	258	244	235	230	248	-2.6%
DUK	683	659	607	609	641	616	505	373	652	695	-1.5%
DTE	532	568	540	558	547	556	542	540	497	488	-1.5%
WEC	763	758	717	684	612	618	613	603	661	644	-1.1%
Mean	541	541	528	532	576	561	539	539	640	652	2.1%
Median	532	533	519	497	547	549	524	540	639	644	1.5%

Source: J.P. Morgan

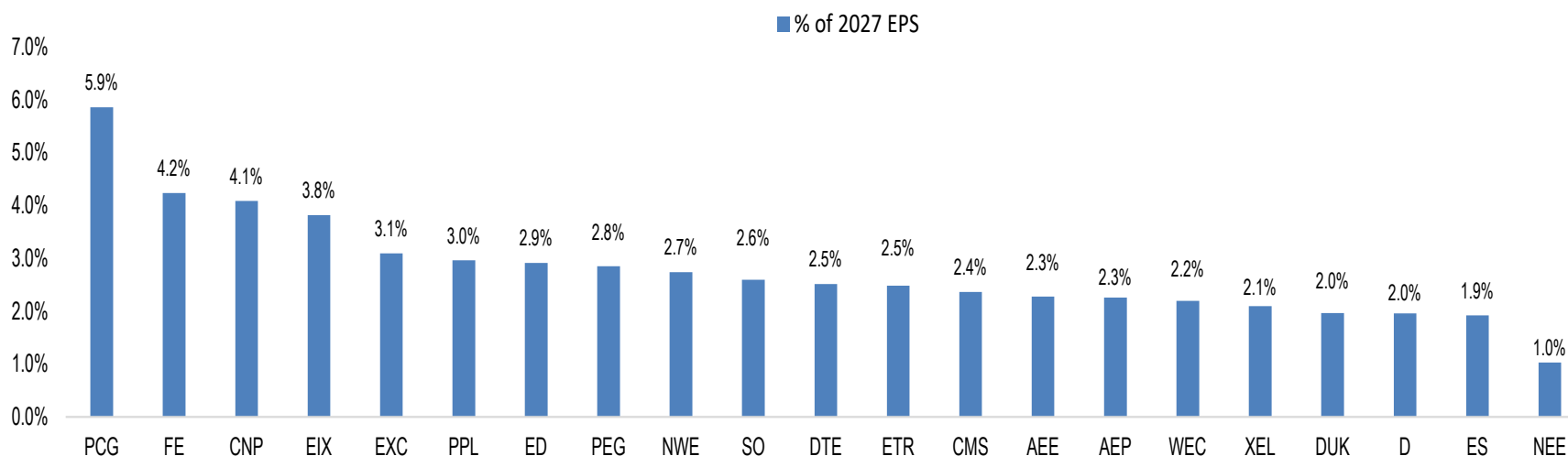
6) Bill Pressure and O&M Analysis: Key Lever to Manage Bill Headroom

Sensitivity Analysis: Load Growth Discussions Muffle Cost Management Focuses

Measuring O&M Sensitivities in Our Utility Coverage Universe

- As load growth drives higher capex plans, O&M management remains a clear tool to protect against adverse regulatory outcomes
- To screen prospective 2027 O&M drag risk, we apply 2.5% cost inflation to 2Q25 TTM O&M
- Among fully regulated utilities, **PCG**, **FE**, **CNP**, **EIX** and **EXC** screen as most exposed
 - Prospective earnings impacts of 3.1% (EXC) to 5.9% (PCG)
 - Though, we acknowledge population growth tailwinds for cost structure stand outside this analysis (notably beneficial for CNP)
- The two CA pureplay IOUs, unsurprisingly, also land at the high end of this screen, with PCG at 5.9% and EIX at 3.8%
 - That said, we acknowledge robust deferrals and other favorable CPUC ratemaking elements provide significant earnings protection
- Conversely, **ES**, **DUK**, and **XEL** stand out with $\leq 2.1\%$ earnings exposure among regulated utilities
 - Versus most of the group banded between 2.2-3.1% (2.5% median)

O&M Sensitivities by Ticker



Source: Bloomberg Finance L.P., J.P. Morgan estimates. Inputs reflect reported GAAP or non-GAAP O&M with minimal exceptions.

7) Regulators & Policy in Focus

Takeaways from NARUC Summer Policy Summit Meetings

Key Takeaways

- Recently [met](#) with >17 state utility commissioners and stakeholders
 - **MS** + surprise; **PA, MO, LA** constructive; **IN** caution
 - **IA, ID, MN** all solid
- Affordability, data center growth, and resource adequacy continued to dominate conversations, building off previous NARUC meetings
- **MS**: hear firm willingness to drive economies of scale and support healthy utility returns in exchange for enabling economic development and strong service quality
- **IN**: Meeting conversations highlighted the potential for the governor's pressure to impact IURC's decisions; we would not be surprised to see incremental downward pressure on authorized ROEs over time
- Affordability top of mind,
 - **DC**: uptick in affordability concerns across the District post recent bill spikes and PJM supply pressures
 - **PA**: affordability pressure point, but independent commission
 - VIU states discussions more positive, with customer protections or benefits in large load growth tariffs or ESA structuring
- While purple state dynamics have represented an area of concern, **MN** and **PA** sounded generally constructive overall and on political risk (and DC as well, for now).

2026 NARUC Winter One-Liner Table

State	Key Takeaways
DC	Affordability has moved firmly to forefront (following calmer backdrop before), while the District appears to have limited levers to reduce bills
ID	Reliability shines as a top priority, with pragmatic lens on growth within least-cost framework and active utility filing cadence; cost economics a swing factor for nuclear
IL	CRGA-enabled framework involves ICC in resource planning; constructive electric engagement persists, persistent ICC ratemaking scrutiny; PJM market reform under debate
IN	Leadership changes and turnover raise concerns over commission independence; affordability report in focus
IA	Welcoming economic development as infrastructure investment opportunity; relatively lower temperature on affordability
LA	Conversations remain supportive of economic development and emphasize customer protections; affordability requires addressing consumption; Cottonwood in crosshairs
MN	Opportunity for large loads done right to bring affordability benefits; relatively lower temp on affordability, recognize rising consumer concerns
MO	Highly constructive regulatory backdrop on display; resource adequacy a priority; solid regulatory relationships for AEE and EVRG, improving regulatory relationship at AQN
MS	Largest positive surprise meeting; economic development welcomed, focus on localities retaining benefits
NM	Resource scarcity reinforces infra needs despite affordability pressures
PA	PUC independence remains intact despite top-down affordability scrutiny; recent AWK order shines as example; see PJM reform as preferred path over structural overhaul

8) Regulatory Risk Ranker

Ranking Relative Risk

OGE, NJR, POR Top Exposure; SWX & PCG Round Out Top Five

- **OGE, NJR, & POR** top the list for rate base exposure due to high single-state concentration
 - OGE: We track a broadly constructive regulatory climate
 - NJR: Believe BGSS/CIP/energy efficiency package can offset rate pressure
 - POR: Beyond GRC, we sense investor attention rests on the POR's ability to secure approval for Holdco structure and WA utility acquisition
- **SWX**: AZ & NV GRC filings serve key near-term regulatory milestones
- **PCG** lands in the top five given its active 2027-30 GRC

ES, EXC, FE, and PEG Also on Our Radar

- See **ES** CL&P's next rate review as the bellwether event for assessing CT's regulatory climate
- For **EXC**, attention resides on active proceedings with relatively lower RRA rankings in MD (~15% rate base) and IL (~25% rate base)
- See risk for **FE** given elevated regional bill pressures and rate cases across OH, WV, and MD, plus an expect 3Q26 NJ filing amid an ongoing BPU evaluation on NJ's ratemaking standards
- See NJ uncertainty as **PEG** plans to file by YE26

Methodology: Tri-factor Ranking

- Our screen encapsulates base size exposure (60% weighting), jurisdictional risk (25% weighting), and time factors (15% weighting)
- We estimate percentage of each utility's rate base undergoing active rate cases or with upcoming applications in 2026. We bracket as follows:
- Jurisdiction/RRA: 1) all Above Average jurisdictions, 2) Average 1 & Average 2, 3) Average 3 & Below Average 1, and 4) Below Average 2 & 3 jurisdictions
- Timing risk: 1) filed in the last 1-3 years as lower risk given recency benefits + having reflected or progressing amid elevated inflation, 2) medium-term cases from 4-5 years ago, and 3) rate cases where 6+ years have elapsed

Reg Risk by Rate Case Size and Time Since Last Filing

Ticker	Blended Score	Rate Base Group	Jurisdiction Score	Duration Score
OGE	3.3	More Than 75%	3.0	Short
NJR	3.3	More Than 75%	3.0	Short
POR	3.3	More Than 75%	3.0	Short
SWX	3.3	More Than 75%	2.7	Short
PCG	3.1	More Than 75%	2.0	Medium
CMS	3.1	More Than 75%	2.0	Short
WEC	2.9	More Than 75%	1.4	Short
MDU	2.7	More Than 50%	2.5	Medium
PEG	2.7	More Than 50%	3.0	Short
AWK	2.6	More Than 50%	2.6	Medium
AEE	2.5	More Than 50%	2.2	Short
ES	2.5	Less Than 50%	3.5	Long
SR	2.5	More Than 50%	2.0	Short
DTE	2.5	More Than 50%	2.0	Short
DUK	2.4	More Than 50%	1.3	Medium
FTS	2.3	Less Than 50%	3.0	Medium
EXC	2.2	Less Than 50%	3.2	Medium
CNP	2.1	Less Than 50%	2.0	Long
FE	2.0	Less Than 50%	2.6	Short
AEP	2.0	Less Than 50%	2.4	Medium
XEL	1.9	Less Than 50%	2.2	Medium
SRE	1.9	Less Than 50%	2.0	Medium
ETR	1.9	Less Than 50%	2.0	Short
ED	1.4	Less Than 10% / Not Active	2.0	Medium
SO	1.4	Less Than 10% / Not Active	2.7	Short
ATO	1.3	Less Than 10% / Not Active	2.0	Short
NI	1.3	Less Than 10% / Not Active	2.0	Short
PPL	0.6	Less Than 10% / Not Active	0.0	N/A
EIX	0.6	Less Than 10% / Not Active	0.0	N/A
EMA	0.6	Less Than 10% / Not Active	0.0	N/A
NWE	0.6	Less Than 10% / Not Active	0.0	N/A

Source: Capital IQ, Company disclosures, J.P. Morgan. Our analysis also includes high probability rate case filings between the date of this publication and year end 2026. Exposure estimates incorporate a combination of company-disclosed and JPM estimated rate base breakdowns, which may differ from actuals.

9) Multiple Expansion vs EPS Revisions in YTD Performance

Attributing YTD Stock Performance to Multiple Expansion vs EPS Revisions

Don't Rein Me In: 2030 Multiple Contraction With Positive EPS Revisions A Recipe For Future Outperformance, In Our View

- We break down YTD share price appreciation between 1) P/E multiple expansion and 2) consensus EPS estimate changes for 2028-2030
- Multiple expansions drive upward performance more than Street EPS revisions in 2028 and 2029, we see this pattern moderating in 2030
 - We see positive EPS revisions both warranting multiple expansion and highlighting a case for future outperformance.
- **AEE** and **ETR** 2030 multiple contractions against growth backdrop could deliver future multiple expansion and/or outperformance
 - See AEE as relatively cleaner for regulatory/political risk; look to soon-to-be-filed IRP as leading indicator
 - Continue to expect hyperscaler clustering across **ETR**'s jurisdictions, creating room for potential upward earnings revisions over time.

ES Multiple Kept Pace Despite EPS Revised Down; NI Potential Catch-up Trade

- **ES** punches above its weight, in our view, with the highest P/E multiple expansion across 2028-2030 (11-13% vs ~1-2% at the group median),
 - But material EPS downward revisions (-5-6% YTD)
- **NI** screens as a catch-up trade, with IN regulatory noise masking GenCo execution
 - See incremental GenCo wins, including a potential MSFT contract, representing pure upside to the 9–10% consolidated EPS CAGR.

9) Multiple Expansion vs EPS Revisions in YTD Performance

Attributing YTD Stock Performance to Multiple Expansion vs EPS Revisions

Utilities Valuation Snapshot

			2028							2029							2030						
	Ticker	Share Price % change YTD	2028 P/E (YE25)	2028 P/E (today)	'28 Multiple Expansion	YTD % price appreciation	% YTD multiple expansion	'28 EPS revisions	2029 P/E (YE25)	2029 P/E (today)	'29 Multiple Expansion YTD	YTD % price appreciation	% YTD multiple expansion	2029 EPS revisions % change YTD	2030 P/E (YE25)	2030 P/E (today)	'30 Multiple Expansion YTD	YTD % price appreciation	% YTD multiple expansion	2030 EPS revisions % change YTD			
Regulated																							
	AEE	7%	16.0x	16.9x	0.9x	7%	6%	1%	14.9x	15.5x	0.7x	7%	5%	2%	14.4x	14.3x	-0.1x	7%	-1%	8%			
	AEP	9%	15.4x	16.6x	1.2x	9%	8%	1%	14.1x	15.2x	1.1x	9%	8%	1%	13.0x	13.8x	0.8x	9%	6%	2%			
	CNP	5%	17.0x	17.6x	0.6x	5%	4%	1%	15.7x	16.2x	0.5x	5%	3%	1%	14.5x	14.9x	0.3x	5%	2%	2%			
	CMS	-2%	15.6x	15.4x	-0.2x	-2%	-2%	0%	14.6x	14.2x	-0.3x	-2%	-2%	1%	14.2x	13.3x	-0.9x	-2%	-7%	5%			
	DTE	6%	14.5x	15.2x	0.7x	6%	5%	1%	13.4x	14.0x	0.6x	6%	5%	2%	12.3x	13.1x	0.8x	6%	7%	0%			
	DUK	3%	15.3x	15.8x	0.5x	3%	3%	0%	14.4x	14.8x	0.4x	3%	3%	0%	13.5x	13.8x	0.3x	3%	2%	1%			
	ED	9%	14.7x	15.7x	1.0x	9%	7%	2%	14.1x	14.6x	0.5x	9%	4%	5%	13.8x	13.8x	0.0x	9%	0%	9%			
	EIX	-1%	8.8x	8.6x	-0.2x	-1%	-2%	1%	8.5x	8.2x	-0.2x	-1%	-3%	2%	8.5x	7.9x	-0.6x	-1%	-7%	6%			
	ES	6%	12.1x	13.5x	1.4x	6%	11%	-5%	11.4x	12.7x	1.3x	6%	11%	-5%	10.7x	12.1x	1.4x	6%	13%	-6%			
	ETR	17%	17.1x	18.8x	1.7x	17%	10%	7%	15.5x	16.8x	1.3x	17%	9%	8%	14.0x	15.2x	1.2x	17%	8%	9%			
	EXC	1%	13.7x	13.5x	-0.2x	1%	-1%	2%	12.9x	12.8x	-0.1x	1%	-1%	2%	12.5x	12.0x	-0.5x	1%	-4%	5%			
	FE	4%	14.3x	14.6x	0.3x	4%	2%	1%	13.2x	13.6x	0.4x	4%	3%	1%	12.4x	12.6x	0.2x	4%	1%	2%			
	NI	0%	17.4x	16.9x	-0.5x	0%	-3%	3%	16.0x	15.3x	-0.7x	0%	-4%	5%	14.8x	14.1x	-0.7x	0%	-5%	5%			
	PCG	-8%	8.2x	7.6x	-0.6x	-8%	-8%	0%	7.5x	7.1x	-0.4x	-8%	-5%	-3%	6.9x	6.5x	-0.4x	-8%	-5%	-3%			
	PPL	0%	15.5x	15.3x	-0.2x	0%	-1%	2%	14.4x	14.2x	-0.1x	0%	-1%	1%	12.9x	13.3x	0.4x	0%	3%	-3%			
	PEG	-8%	15.9x	14.8x	-1.2x	-8%	-7%	0%	14.9x	13.9x	-1.0x	-8%	-7%	-1%	13.7x	13.2x	-0.5x	-8%	-4%	-4%			
	SO	2%	16.6x	16.6x	0.0x	2%	0%	2%	15.6x	15.5x	-0.1x	2%	0%	2%	14.4x	14.4x	0.0x	2%	0%	2%			
	WEC	2%	16.2x	16.5x	0.3x	2%	2%	0%	15.1x	15.3x	0.3x	2%	2%	0%	14.0x	14.1x	0.2x	2%	1%	0%			
	XEL	4%	15.1x	15.4x	0.3x	4%	2%	2%	13.9x	14.3x	0.3x	4%	2%	2%	12.8x	13.3x	0.6x	4%	4%	0%			
	Mean		14.7x	15.2x	0.6x	5%	4%	1%	13.7x	14.1x	0.4x	5%	3%	1%	13.1x	13.1x	0.1x	5%	0%	2%			
	Median		15.4x	15.7x	0.7x	6%	4%	1%	14.2x	14.7x	0.5x	6%	4%	1%	13.7x	13.8x	0.1x	6%	1%	2%			
	Mean (ex-EIX, PCG)		15.4x	15.8x	0.4x	4%	3%	1%	14.4x	14.7x	0.3x	4%	2%	2%	13.4x	13.6x	0.2x	4%	2%	2%			
	Median (ex-EIX, PCG)		15.5x	15.7x	0.3x	4%	2%	1%	14.4x	14.6x	0.4x	4%	3%	1%	13.7x	13.8x	0.2x	4%	1%	2%			
SMID																							
	MDU	2%	16.7x	17.0x	0.3x	2%	2%	0%	15.1x	15.7x	0.6x	2%	4%	-3%	15.3x	14.0x	-1.3x	2%	-8%	9%			
	NWE	11%	15.4x	16.9x	1.5x	11%	10%	1%	14.6x	16.2x	1.5x	11%	10%	1%	13.0x	15.3x	2.3x	11%	18%	-5%			
	OGE	11%	15.3x	16.8x	1.5x	11%	10%	1%	14.3x	15.6x	1.4x	11%	10%	1%	13.3x	14.6x	1.3x	11%	9%	1%			
	POR	3%	12.8x	13.0x	0.2x	3%	2%	1%	12.2x	12.6x	0.4x	3%	3%	0%	12.6x	11.4x	-1.1x	3%	-9%	13%			
	Mean		15.0x	15.9x	0.9x	7%	6%	1%	14.1x	15.0x	1.0x	7%	7%	0%	13.5x	13.8x	0.3x	7%	2%	4%			
	Median		15.3x	16.8x	0.9x	7%	6%	1%	14.5x	15.7x	1.0x	7%	7%	0%	13.2x	14.3x	0.1x	7%	1%	5%			
Diversified																							
	SRE	-3%	14.8x	14.2x	-0.6x	-3%	-4%	1%	13.7x	13.0x	-0.7x	-3%	-5%	2%	12.8x	11.9x	-0.9x	-3%	-7%	4%			
	Mean (Regulated & Diversified)		14.7x	15.0x	0.3x	3%	2%	1%	13.7x	13.9x	0.2x	3%	1%	1%	12.8x	12.9x	0.1x	3%	0%	2%			
	Median (Regulated & Diversified)		15.4x	15.4x	0.3x	3%	2%	1%	14.2x	14.3x	0.3x	3%	2%	1%	13.3x	13.3x	0.1x	3%	1%	2%			
LDCs																							
	ATO	0%	17.5x	17.1x	-0.4x	0%	-2%	2.6%	16.2x	15.9x	-0.3x	0%	-2%	2%	15.3x	14.8x	-0.5x	0%	-3%	3%			
	NJR	17%	12.3x	14.3x	2.0x	17%	16%	-0.2%	11.4x	13.2x	1.7x	17%	15%	2%	10.7x	12.5x	1.8x	17%	17%	0%			
	SR	1%	13.4x	13.6x	0.3x	1%	2%	-1%	13.0x	12.7x	-0.2x	1%	-2%	3%	12.3x	12.0x	-0.3x	1%	-3%	4%			
	SWX	11%	13.9x	15.1x	1.2x	11%	8%	2.1%	12.9x	13.3x	0.4x	11%	3%	8%	13.4x	12.3x	-1.2x	11%	-9%	12%			
	Mean		14.3x	15.0x	0.8x	7%	6%	1%	13.4x	13.8x	0.4x	7%	4%	4%	12.9x	12.9x	0.0x	7%	1%	5%			
	Median		13.6x	14.7x	0.7x	6%	5%	1%	12.9x	13.2x	0.1x	6%	1%	2%	12.9x	12.4x	-0.4x	6%	-3%	4%			
Canadians																							
	AQN	-7%	13.3x	13.0x	-0.3x	-7%	-2%	-5.4%	12.8x	12.5x	-0.4x	-7%	-3%	-5%	11.8x	11.6x	-0.2x	-7%	-2%	-6%			
	EMA	4%	17.2x	17.9x	0.7x	4%	4%	-0.3%	16.2x	16.6x	0.4x	4%	3%	1%	15.2x	15.5x	0.2x	4%	2%	2%			
	FTS	7%	17.5x	18.6x	1.1x	7%	7%	1%	16.3x	17.4x	1.1x	7%	7%	1%	16.1x	16.5x	0.4x	7%	3%	5%			
	Mean		16.0x	16.5x	0.5x	1%	3%	-2%	14.3x	14.7x	0.4x	1%	2%	-1%	13.8x	13.8x	0.2x	1%	1%	0%			
	Median		17.2x	17.9x	0.7x	4%	4%	0%	13.4x	13.8x	0.4x	4%	3%	1%	12.9x	12.9x	0.2x	4%	2%	2%			

Source: Bloomberg L.P.; Prices as of 9/8/2026. Please see our accompanying note [here](#)

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9) Multiple Expansion vs EPS Revisions in YTD Performance

Attributing YTD Stock Performance to Multiple Expansion vs EPS Revisions

Utilities Ranked by % 2028 and 2029 P/E Multiple Expansions (highest to lowest)

2028 Ranked					2029 Ranked					2030 Ranked				
Ticker	28 Multiple Expansion YTD	YTD % price appreciation	% YTD multiple expansion	'28 EPS revisions	Ticker	29 Multiple Expansion YTD	YTD % price appreciation	% YTD multiple expansion	2029 EPS revisions % change YTD	Ticker	29 Multiple Expansion YTD	YTD % price appreciation	% YTD multiple expansion	2030 EPS revisions % change YTD
NJR	2.0x	17%	16%	0%	NJR	1.7x	17%	15%	2%	NWE	2.3x	11%	18%	-5%
ES	1.4x	6%	11%	-5%	ES	1.3x	6%	11%	-5%	NJR	1.8x	17%	17%	0%
ETR	1.7x	17%	10%	7%	NWE	1.5x	11%	10%	1%	ES	1.4x	6%	13%	-6%
OGE	1.5x	11%	10%	1%	OGE	1.4x	11%	10%	1%	OGE	1.3x	11%	9%	1%
NWE	1.5x	11%	10%	1%	ETR	1.3x	17%	9%	8%	ETR	1.2x	17%	8%	9%
SWX	1.2x	11%	8%	2%	AEP	1.1x	9%	8%	1%	DTE	0.8x	6%	7%	0%
AEP	1.2x	9%	8%	1%	FTS	1.1x	7%	7%	1%	AEP	0.8x	9%	6%	2%
ED	1.0x	9%	7%	2%	DTE	0.6x	6%	5%	2%	XEL	0.6x	4%	4%	0%
FTS	1.1x	7%	7%	1%	AEE	0.7x	7%	5%	2%	PPL	0.4x	0%	3%	-3%
AEE	0.9x	7%	6%	1%	MDU	0.6x	2%	4%	-3%	FTS	0.4x	7%	3%	5%
DTE	0.7x	6%	5%	1%	ED	0.5x	9%	4%	5%	CNP	0.3x	5%	2%	2%
EMA	0.7x	4%	4%	0%	CNP	0.5x	5%	3%	1%	DUK	0.3x	3%	2%	1%
CNP	0.6x	5%	4%	1%	POR	0.4x	3%	3%	0%	EMA	0.2x	4%	2%	2%
DUK	0.5x	3%	3%	0%	SWX	0.4x	11%	3%	8%	FE	0.2x	4%	1%	2%
FE	0.3x	4%	2%	1%	DUK	0.4x	3%	3%	0%	WEC	0.2x	2%	1%	0%
SR	0.3x	1%	2%	-1%	FE	0.4x	4%	3%	1%	SO	0.0x	2%	0%	2%
XEL	0.3x	4%	2%	2%	EMA	0.4x	4%	3%	1%	ED	0.0x	9%	0%	9%
POR	0.2x	3%	2%	1%	XEL	0.3x	4%	2%	2%	AEE	-0.1x	7%	-1%	8%
MDU	0.3x	2%	2%	0%	WEC	0.3x	2%	2%	0%	AQN	-0.2x	-7%	-2%	-6%
WEC	0.3x	2%	2%	0%	SO	-0.1x	2%	0%	2%	SR	-0.3x	1%	-3%	4%
SO	0.0x	2%	0%	2%	PPL	-0.1x	0%	-1%	1%	ATO	-0.5x	0%	-3%	3%
EXC	-0.2x	1%	-1%	2%	EXC	-0.1x	1%	-1%	2%	PEG	-0.5x	-8%	-4%	-4%
PPL	-0.2x	0%	-1%	2%	ATO	-0.3x	0%	-2%	2%	EXC	-0.5x	1%	-4%	5%
CMS	-0.2x	-2%	-2%	0%	SR	-0.2x	1%	-2%	3%	NI	-0.7x	0%	-5%	5%
EIX	-0.2x	-1%	-2%	1%	CMS	-0.3x	-2%	-2%	1%	PCG	-0.4x	-8%	-5%	-3%
ATO	-0.4x	0%	-2%	3%	EIX	-0.2x	-1%	-3%	2%	CMS	-0.9x	-2%	-7%	5%
AQN	-0.3x	-7%	-2%	-5%	AQN	-0.4x	-7%	-3%	-5%	EIX	-0.6x	-1%	-7%	6%
NI	-0.5x	0%	-3%	3%	NI	-0.7x	0%	-4%	5%	SRE	-0.9x	-3%	-7%	4%
SRE	-0.6x	-3%	-4%	1%	SRE	-0.7x	-3%	-5%	2%	MDU	-1.3x	2%	-8%	9%
PEG	-1.2x	-8%	-7%	0%	PCG	-0.4x	-8%	-5%	-3%	SWX	-1.2x	11%	-9%	12%
PCG	-0.6x	-8%	-8%	0%	PEG	-1.0x	-8%	-7%	-1%	POR	-1.1x	3%	-9%	13%

Source: Bloomberg L.P.; Prices as of 9/8/2026. Please see our accompanying note [here](#)

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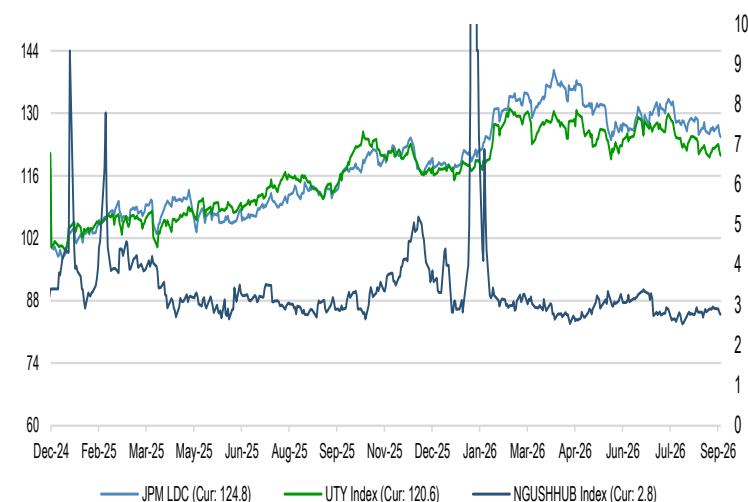
10) Gas Views

Thoughts Across LDCs & Canadians

Positive on LDCs; Prefer U.S. over Canada

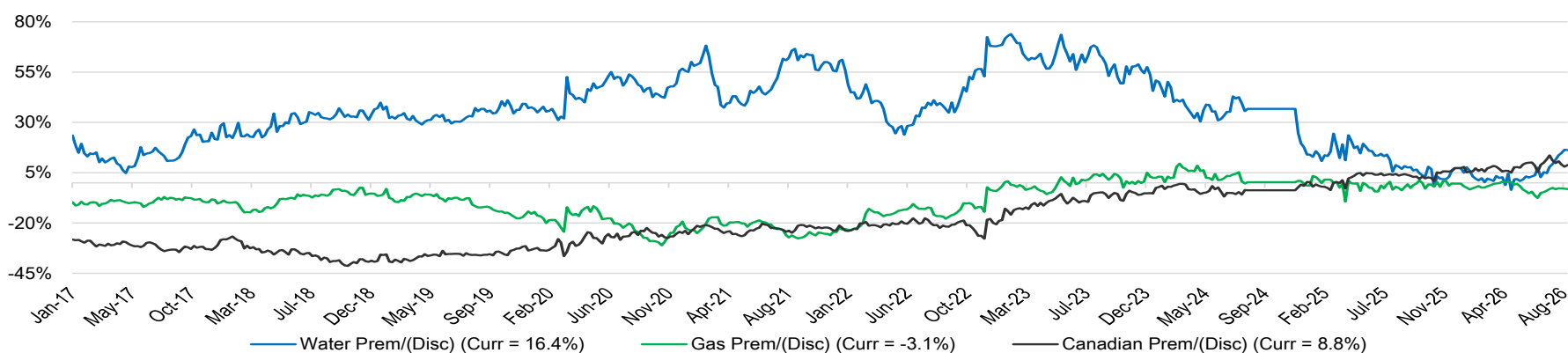
- **LDCs:** we remain positive on LDCs overall and expect continued strong group interest linked to 1) a growing recognition that capital tailwinds also exist for the LDCs broadly, and 2) increasingly strong gas sentiment
 - **NI:** GenCo positions company to capture superior returns on robust DC gen demand
 - **NJR:** S&T projects stand out; visible CEV growth; NJNG a reliable growth engine
 - **SWX:** NV & AZ progress toward formula rates; GB expansions add to earnings profile
 - **ATO:** leading rate base growth + history of earnings outperformance VS fuller valuation
 - **SR:** MO rate case + TN integration support re-base; execution critical for re-rating
- **CAD:** maintain US over CAD relative preference given US utility generally offer higher growth with greater line of sight and protection via more balance sheet cushion
 - **EMA:** Tampa powers visibility; Nova Scotia backdrop evolving; await DC contracting
 - **AQN:** Pure play regulated pivot underway; critical rate case execution phase ongoing
 - **FTS:** AZ large load & formula rates topical; MISO LRTP + BC LNG drive capex upside

2Y Sector Price (Normalized) versus Henry Hub NG



Source: J.P. Morgan estimates, Bloomberg Finance LP. Prices as of 9/10/2026.

Sector Relative Valuation vs JPM Regulated Electric Utilities



Source: J.P. Morgan estimates, Bloomberg Finance LP. Prices as of 9/10/2026 close.

Appendix



Key Recent Reports

Policy & Corporate Updates: Recent Access Events

Expert Access

- [ACC Discussion 8/24/26](#): Constructive Again; Finding the Balance Between Data Center Demand and Affordability
- [NARUC Summer Summit Takeaways, 7/23/26](#): Industry and state specific takeaways from >17 Commissioner Meetings we hosted in Minneapolis
- [PAPUC Discussion 5/1/26](#): The State of Virtue, Liberty and Commission Independence; No Finger on the Scale Reaffirmed
- [NARUC 2026 Winter Summit Takeaways, 2/19/26](#): Industry and state specific takeaways from >20 Commissioner Meetings we hosted in D.C.
- [PUCT Discussion, 3/16/26](#): TX Miracle Shines Through; Batch Study Process Avoids Analysis Paralysis, Positive Meeting
- [GA PSC Discussion, 3/30/26](#): Bill affordability, economic development and mobility, regulatory benchmarks, and rate/capital structure in focus
- [CO PUC Discussion, 11/05/25](#): Affordability; Tariff design for load growth; Strategic location choices; Resource mix; Wildfire backdrop
- **Data Center Series**: [McKinsey, 6/17/2025](#): Data centers, new project locations, fuel mix, regulatory outlooks; [JPM Power Trader, 1/2/2025](#): Power supply-demand, FTM opportunities, PJM uncertainty; [LineVision, 6/11/2024](#): Capital-light solution, complements utility transmission investment
- **Nuclear Energy Series Takeaways**: [Part XI, 7/21/25](#): Kairos Power, NuScale Power, TerraPower, Oklo; [Part X, 10/28/24](#): Fireside chat with Holtec International, [Part IX, 7/29/24](#): Fireside chats with X-energy, Westinghouse, TerraPower; [Part VIII, 4/22/24](#): Fireside chats with Ontario Power Generation and Terrestrial Energy; [Part VII, 9/13/23](#): Fireside chat with TerraPower; [Part VI, 7/6/23](#): Oklo and GE; [Part V, 9/28/22](#): Fireside chats with GE, BWXT, and Terrestrial Energy; [Part IV, 6/12/22](#): TerraPower, DOE Loan Programs, and Ontario Power Generation; [Part III, 4/27/22](#): MIT, X-Energy, and Westinghouse; [Part II, 3/31/22](#): Pillsbury, Oklo, PSEG; [Part I, 1/31/22](#): DOE Office of Nuclear Energy, CATF, and NuScale Power.
- **CCUS Series Takeaways**: [JPM Carbon Trading 10/28/2024](#); [CCUS Compilation of 2023 Firesides, 10/14/2023](#): Fireside chats with Summit, Wolf, Navigator; [CCUS Compilation of 2022 Firesides, 6/19/2022](#): Fireside chats with LA DNR, Summit, Wolf, Carbon Solutions, Navigator; [CCUS Compilation of 12 Fireside Chat, 5/30/22](#): Fireside chats with DOE, IEA, 8 Rivers, Navigator, Summit, GaffneyCline, CATF, Columbia University, Rice University, University of Houston, & CEM. [Part 1, 6/2/22](#); [Part 2, 6/17/22](#); and [Part 3, 7/27/22](#).

Corporate Access

- [JPM Natural Resources Conference Takeaways, 6/25/26](#): We cover industry-wide takeaways as well as compile our company specific takeaways from the J.P Morgan Natural Resources Conference
- [AGA 2026 Takeaways, 5/20/26](#): Takeaways from our 21 company meetings at the AGA Financial Forum in Scottsdale, AZ. Question bank [here](#)
- [JPM New England Utilities Mini-Forum Takeaways, 1/23/26](#): Takeaways from our company meetings at the JPM New England Forum; Question bank [here](#)
- [EEI Conference 2025 Takeaways, 11/13/25](#): Takeaways from our 28 company meetings at the 2025 EEI Financial Forum; Question bank [here](#).
- [JPM Rocky Mountain Utilities Forum Takeaways, 3/10/25](#): Takeaways from our company meetings at the JPM Rocky Mountain Forum; Question bank [here](#)
- **NDRs with**: [AES](#), [NI](#), [ETR](#), [CMS](#), [SRE](#), [EIX](#), [DTE](#), [D](#), [CNP](#), [WEC](#).

Drawn & Quartiled: Navigating Election, Affordability & Rate Cases

Utilities with Highest Exposure to Election and Rate Case Risks

NEE & ETR Among Lower Risk; EXC and ES Screen Most at Risk

- Leveraging our recent [regulatory risk ranker](#) and [bill pressure](#) analyses, we screen our coverage universe across these factors as well as 2026 gubernatorial election exposure and jurisdiction ratings (per RRA)
- Overall, we highlight relatively favorable positions for **NEE** and **ETR**, potentially less **XEL** risk than bears believe, while **EXC** and **ES** screen higher risk

2026 Gubernatorial Election Risk Ranker

	Blended Score	Rate Base Exposure to 2026 Gubernatorial Elections
SO	3.3	Quartile 4
SWX	3.2	Quartile 4
ED	3.2	Quartile 4
POR	3.2	Quartile 4
EXC	3.2	Quartile 3
CMS	3.0	Quartile 4
DTE	3.0	Quartile 4
XEL	2.6	Quartile 4
ATO	2.6	Quartile 3
AEE	2.6	Quartile 1 / Not Active
CNP	2.6	Quartile 3
ES	2.5	Quartile 2
WEC	2.4	Quartile 3
D	2.4	Quartile 1 / Not Active
AES	2.4	Quartile 1 / Not Active
NEE	2.4	Quartile 4
EIX	2.4	Quartile 3
PCG	2.4	Quartile 3
NI	2.3	Quartile 2
AEP	2.3	Quartile 2
FE	2.3	Quartile 2
AWK	2.1	Quartile 2
DUK	2.1	Quartile 2
ETR	2.1	Quartile 2
PPL	1.8	Quartile 1 / Not Active
NWE	1.4	Quartile 1 / Not Active

Utilities with Highest Exposure to Election and Rate Case Risks

	Blended Score	Rate Base Exposure to Active Elections and Rate Cases
EXC	3.5	Quartile 3
SWX	3.2	Quartile 4
AES	3.2	Quartile 3
AEE	3.0	Quartile 2
CMS	3.0	Quartile 4
DTE	3.0	Quartile 4
WEC	2.9	Quartile 4
FE	2.9	Quartile 2
ES	2.8	Quartile 2
PCG	2.8	Quartile 4
XEL	2.6	Quartile 4
SO	2.6	Quartile 1
PPL	2.6	Quartile 3
ETR	2.4	Quartile 3
AEP	2.0	Quartile 1
NI	1.8	Quartile 2
ATO	1.6	Quartile 1
AWK	1.6	Quartile 1

Source: Capital IQ, EIA, BEA, company disclosures, J.P. Morgan. Please see our accompanying note [here](#). In the left hand table, we evaluate risk exposure for utilities operating in states with upcoming gubernatorial elections, weighting each company by rate base exposure, affordability direction (wallet share and total bill), and jurisdictional risk. On the right, we take this analysis one step further by considering rate base exposed to active rate cases.

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Drawn & Quartiled: Navigating Election, Affordability & Rate Cases

Screening Our Coverage Across Regulatory, Election, Rate Case, and Affordability Risks

NEE & ETR Among Lower Risk; EXC, ES Screen Most at Risk

- Leveraging our recent [regulatory risk ranker](#) and [bill pressure](#) analyses, we screen our coverage universe across these factors as well as 2026 gubernatorial election exposure and jurisdiction ratings (per RRA)
- Here, we analyze which utilities could face greater regulatory or public scrutiny heading into 2026
- Overall, we highlight relatively favorable positions for **NEE** and **ETR**
 - **NEE** screens favorably with: better-than-median affordability metrics and a high quality regulatory environment in FL
- While **EXC** and **ES** screen higher risk
 - **EXC** faces 1) multiple state elections, 2) comparatively restrictive regulatory climates, 3) high YoY rate of change in utility bills
 - **ES** screens unfavorably given challenged regulatory jurisdiction and high wallet share

Methodology

- We split weight equally between state stakeholder scores (regulatory and election risk) and affordability metrics while leaving a smaller weight to the RRA score. (higher value implies higher risk)
- We leverage proprietary composite scores from our [regulatory risk ranker](#) and election risk ranker
- Affordability: we bucket states into quartiles, with those in the most improved group scoring 1 and most deteriorated scoring a 4, and weight by rate base per state to arrive at a weighted-average company score
- RRA: brackets: 1) all Above Average jurisdictions, 2) Average 1 and Average 2, 3) Average 3 and Below Average 1, and 4) Below Average 2 and 3 jurisdictions

Composition Ranker

	Jursidiction Score	Election Ranker Score	Regulatory Risk Ranker Score	Wallet Share Score	Wallet Share Change Score	Bill Direction Score	Overall Score
EIX	1	2	1	1	1	1	1.2
NEE	1	2	1	1	1	1	1.2
NWE	4	1	1	1	1	1	1.3
PCG	1	2	4	1	1	1	1.9
D	3	2	1	2	3	3	2.2
ETR	3	1	4	2	1	2	2.2
NJR	4	1	1	1	4	4	2.2
PEG	4	1	1	1	4	4	2.2
WEC	1	3	3	1	2	2	2.2
XEL	2	3	4	1	1	1	2.2
POR	1	4	1	1	3	3	2.3
ATO	3	3	2	3	1	2	2.3
CNP	3	3	2	2	2	2	2.3
PPL	1	1	3	4	2	3	2.4
DUK	1	1	2	4	3	4	2.4
AWK	3	1	3	3	3	3	2.6
AEP	3	2	3	3	2	3	2.6
AEE	4	3	3	2	2	2	2.7
CMS	1	4	3	3	3	1	2.7
NI	1	2	2	4	4	4	2.8
FE	3	2	2	4	4	3	2.9
ED	2	4	2	3	3	3	2.9
DTE	1	4	4	3	3	1	3.0
ES	4	3	4	4	2	2	3.2
SO	1	4	2	4	4	4	3.3
AES	1	2	4	4	4	4	3.3
SWX	4	4	4	3	2	2	3.3
EXC	4	4	3	2	4	4	3.5

Source: Capital IQ, EIA, BEA, company disclosures, J.P. Morgan. Weightings 10% RRA, 45% election + regulatory, 45% affordability. Please see accompanying note [here](#)

Rate Base vs Capital Plan Rolls

Implied Five-Year Capital Plan Roll Forward and Percentage Increase

Ticker	Current 5-YR Capex Plan	Required Roll- Forward Plan	Implied Increase %
POR	\$8	\$7 - \$7	(6.5%)
EXC	\$42	\$39 - \$41	(4.0%)
EIX	\$40	\$37 - \$39	(2.8%)
AEP	\$78	\$77 - \$81	0.8%
SWX	\$6	\$6 - \$7	1.4%
FTS	\$29	\$29 - \$31	3.3%
ED	\$38	\$39 - \$41	6.5%
SR	\$5	\$5 - \$5	6.9%
PEG	\$24	\$25 - \$26	7.1%
FE	\$36	\$38 - \$40	7.8%
MDU	\$3	\$3 - \$3	8.1%
PPL	\$23	\$24 - \$26	9.0%
DUK	\$103	\$110 - \$115	9.1%
CNP	\$34	\$36 - \$38	9.4%
SRE	\$65	\$70 - \$73	10.2%
OGE	\$7	\$8 - \$8	10.3%
SO	\$81	\$87 - \$92	10.4%
AEE	\$32	\$35 - \$36	11.4%
EMA	\$20	\$22 - \$23	11.8%
WEC	\$38	\$41 - \$43	12.9%
XEL	\$60	\$67 - \$71	14.7%
ATO	\$26	\$29 - \$31	15.5%
CMS	\$24	\$28 - \$30	21.3%



EEI 2025 Takeaways

Growth 6, 7%? Nah Bruh 7-9% - Gen Cooking

Ticker	EEI One Liners
AEE	Messaging confidence in EPS growth - AEE not to constrain EPS growth if investments point higher; data center signed construction agreements now total ~3GW; attention on IL regulatory environment
AEP	Highlighting clear confidence, as well as conservatism, in the 2026-2030 outlook, with large load upward bias; marching along earned ROE improvement path; see benefit in 11-state footprint
AES	Signaling upward bias to prevailing growth rate ahead of 4Q plan refresh; IN data center deal appears incoming, with focus on resource requirements; expect OH MYP filing in short order
AQN	Rate case execution across Empire Electric, CalPeco, and New England critical for improved stakeholder support and earnings improvement; Balanced approach to portfolio optimization, await refreshed business outlook
AWK	Full heart, clear eyes on the WTRG acquisition; acknowledged potential for medium-term earnings growth above the range outlook depending on strategic LDC evaluation, though do not expect change in growth messaging
CMS	1GW battery step up to benefit FCM; does growth CAGR move up?; 1-2GW of data center opportunities sprinting to the finish line, with another 4-5GW jogging along; Campbell likely in the picture for foreseeable future
CNP	Plan, execute, enhance: capital upside opportunities remain; diverse load growth drivers emphasized, data center announcements likely to come; Job's not done: TX legislation priorities would aim to chip away at 20bps of reg lag
D	First power 2H of 1Q26; focus remains on Charybdis following delay announced on 3Q call; data centers: the same slice of a bigger pie; 4Q plan roll to highlight gen capex stepping up, but dividend growth still a bit away
DTE	Positive tone on additional hyperscaler deals as demand recovery convo shifts; expect each deal to feature bespoke terms; gearing up for next gas filing and increasing electric IRM in focus; Vantage unicorn(s)
DUK	Solid load growth outlook w/ 3GW data center ESAs, telegraphed \$8bn base plan increase largely for generation, \$10bn upside in firming up cost estimates and economic development opportunities; capex step up expected with roll
ED	Positive tone on CECONY joint proposal; expect downward shift in equity needs; due to surcharges, earnings investment base more accurate measure than rate base; cautious optimism on regulated renewables potential
EIX	Reiterating confidence in 5-7% growth ahead of CoC proceeding and 2026 EPS guide; SB 254 Phase II dwells in spotlight; no update on Eaton investigation; transmission outlook in focus
EMA	Tampa Electric's 8-9% rate base CAGR + new rates + upside investment opportunities drive earnings; evolving regulatory landscape critical to Nova Scotia outlook; NMGC divestment execution to drive increased capital flexibility
ES	PURA's new groove takes the spotlight; NSTAR Gas reconsideration confidence, more MA gas pipeline possible; balance sheet improvement plan striding along; OSW inching closer to completion
ETR	Industrial sales CAGR higher on strong, yet conservative, load growth outlook; upside opportunity remaining; dispatchable generation growth may drive incremental capital investment within the current plan period
EXC	Distilling a three-step gameplan to address PJM's market challenges; ComEd MYP reconciliation in focus alongside MD rate cases
FE	WV gas plant opportunity leading the charge for generation upside as FE parses through EPC vs BoT... coupled with robust transmission growth outlook; OH clarity top of mind
FTS	Incremental large load in AZ + transmission opportunities + BC LNG drive upside to refreshed \$28.8bn capital plan; AZ regulatory momentum on track, full run-rate in 2027; no discrete equity needs in unchanged funding profile
NEE	w/ BYOP appetite increasing, large-scale + full-service model provides meaningful advantages; renewed nuclear interest driven by combo of power demand and customer interest; labor tightness a top consideration for gas builds
NI	Among strongest EEI meetings; GenCo speed to market / return profile stand out; drive upside to base business; 3 GW contract headed for IURC approval 1H26; \$1bn+ customer savings eases IN bill pressure
NWE	All I want for Christmas is an ESA (hopeful for Sabey by YE25); pending SD sales tax exemption seen as triggering event for more activity; BKH merger process alive and well
PCG	SB 254 Phase II remains in the early innings; abstract filings cover the gamut of solutions ahead of deeper analysis; capital allocation priorities center on capex plan; focus on solid YTD ignition reductions
PEG	ZEC repayments issued to offer near-term customer rate relief, with eyes on longer term IRP and nuclear solutions; no incremental updates on nuclear contracting
POR	All eyes on regulatory execution and bill affordability into 2026; holdco reorg resonates as key lever to offset size of equity needs; continued OR commitment to renewables
PPL	See PA regulated utility ownership as backstop solution to PPL's genco JV; expecting KY rate case approval by YE25; new load may trigger incremental generation to latest KY CPCN
SO	Earnings rebase opportunity front and center, with 2028 resonating as the most likely base; regulatory agenda screens sparse near term in focus post PCS elections
WEC	VLC tariff decision upcoming w/ strong ROE, equity layer; purchase cancellation payment agreements enabled earlier work; IL gas pipeline investment to ramp higher in 27-29; load growth jumps forward, including 1.3GW from Vantage
XEL	3GW base / >20GW data center pipeline represent sales and rate base opportunity; wildfire liabilities mostly wound down, post Marshall settlement and material Smokehouse Creek progress

Source: J.P. Morgan; Please see accompanying note [here](#)

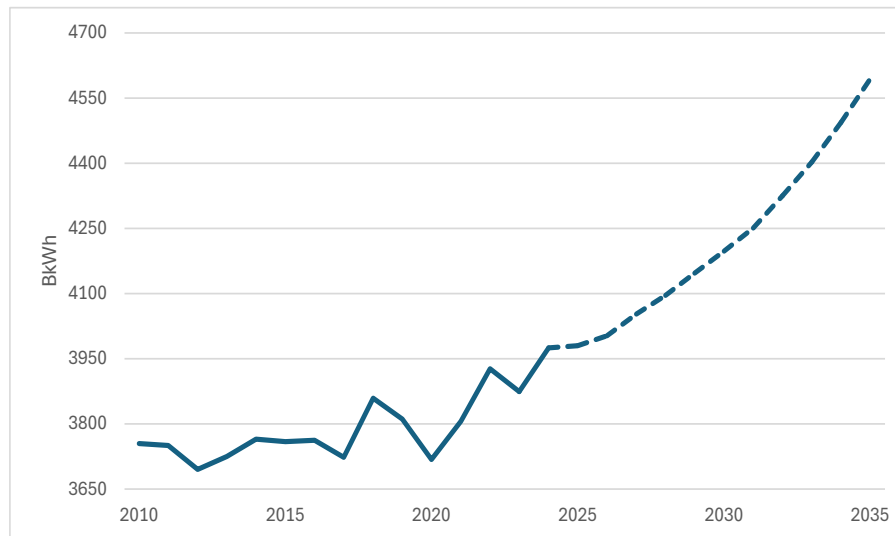
Don't Stop Believing, Hold On To That Power Feeling

Once-in-a-Generation Power Demand Growth Opportunity

Secular Demand Tailwinds Favor Incumbent IPP Position

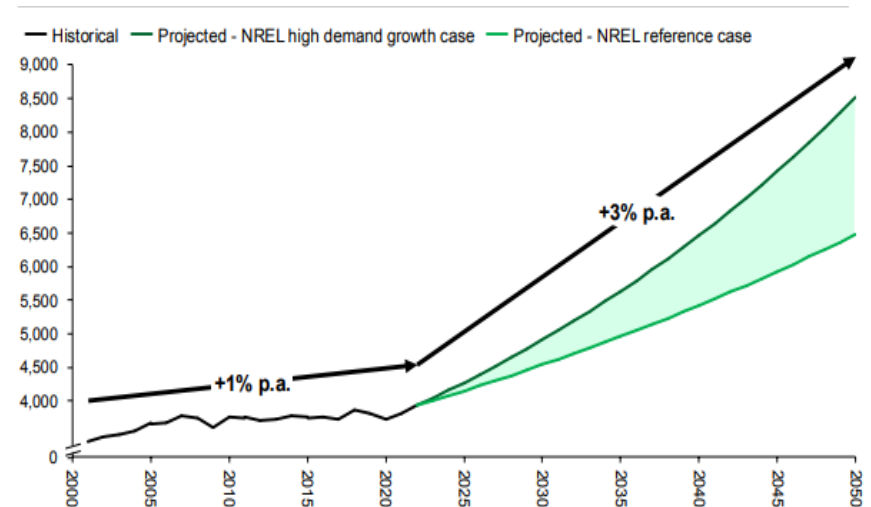
- Paradigm shift in demand driven by
 - Manufacturing onshoring, broader electrification trends (transportation, heating, and more), data center development
- Structural demand shift + limited dispatchable resource additions = outsized margin for IPPs.
- Do not see resource supply developing in a fashion to match the pace of demand growth
 - Fueling appetite for contracting and/or overall higher prices to benefit IPPs
- EIA's [projects](#) U.S. annual electricity consumption increasing in 2025 and 2026, surpassing the all-time high reached in 2024
 - With growth driven by the commercial sector (which includes data centers), and the industrial sector (which includes manufacturing reshoring)
- The DOE's September 2024 Advanced Nuclear report notes that electricity demand could more than double by 2050

U.S. Electricity Demand (2010-2035)



Source: EIA

U.S. Electricity Demand (2010-2035)



Source: DOE

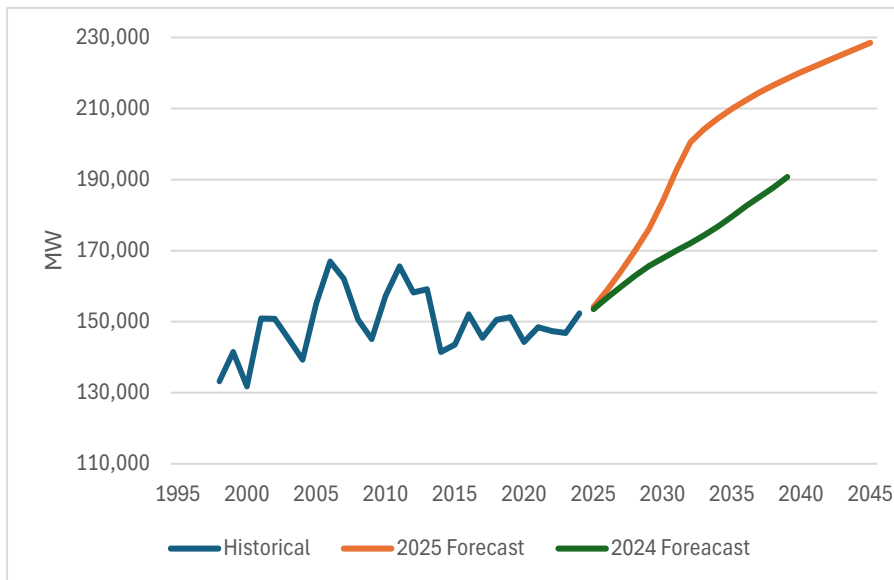
Don't Stop Believing, Hold On To That Power Feeling

Electrification, Onshoring and Data Centers Drive Load Growth Inflection Point

Power Prices Moving Up; Fleet Utilization Increasing

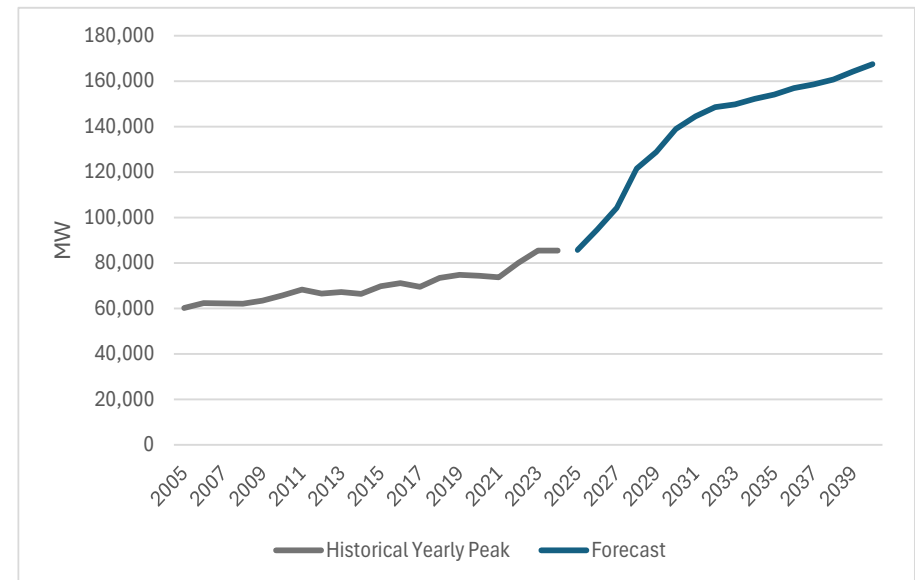
- Forward pricing have increased in both PJM and ERCOT
 - PJM West and East 2026 ATC power prices have risen ~\$6/MWh YTD, with 2027 ATC power prices up ~\$4/MWh
 - ERCOT 2027 ATC prices have stepped up ~\$5/MWh YTD
- PJM's 2025 Long-Term Load Forecast Report highlighted a clear upward bias in load growth forecasts versus 2024 estimates
 - PJM [projects](#) net energy load growth to average 4.8% annually over the next 10-year period, and 2.9% over the next 20 years
- ERCOT [forecasts](#) ~145GW of peak summer load by 2031, representing a ~59GW increase versus the ~85GW forecasted 2025 levels
 - Data centers drive the largest portion of growth, particularly with a substantial uplift in demand occurring in 2028
 - Crypto, Industrial, and Hydrogen represent the next largest demand sources

PJM Summer Peak Load Growth Forecasts



Source: PJM, estimates as of January 2025

ERCOT Historical and Forecast Peak Demand



Source: ERCOT, April 8, 2025

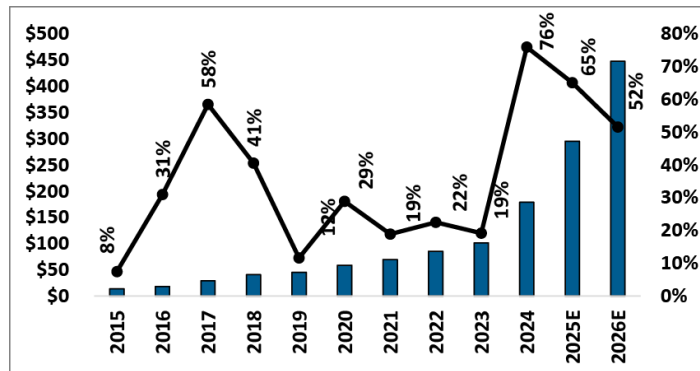
Don't Stop Believing, Hold On To That Power Feeling

Data Centers Power Demand Growth Remains Robust

Data Center Development Remains Strong

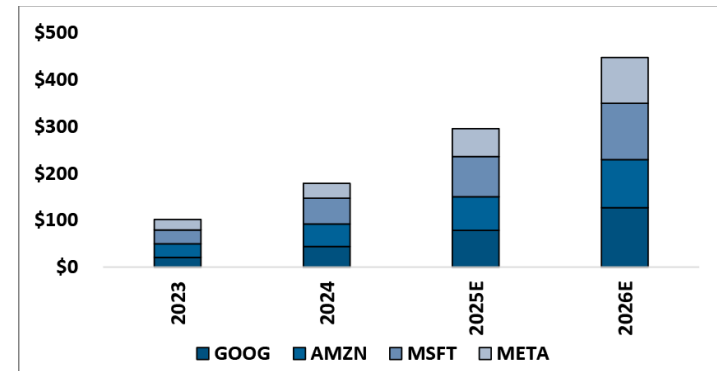
- Data centers = 4.4% of total U.S. electricity in 2023, growing to consume approximately 6.7-12% of total U.S. electricity by 2028
- JPM Hardware, Telecom & Networking Equipment team's [sees](#) \$150bn+ 2026 capex growth among AMZN, GOOG, MSFT, META in 2026
- JPM Communications Infrastructure team [estimates](#) 2025 total U.S. data center supply capacity at 34.5 GW, up to 74.5 GW by the end of 2028, versus 43.4 GW of demand in 2025, up to 105.1 GW in 2028
- Positive hyperscaler data points + tight supply/demand fundamentals = healthy runway for commercial opportunities across the IPP fleet

Annual Data Center Capex from Top 4 U.S. CSPs



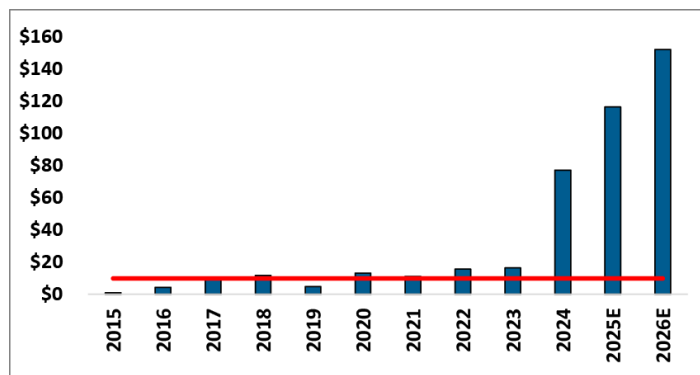
Source: 650 Group and J.P. Morgan Hardware and Networking estimates

Annual Data Center Capex from Top 4 U.S. CSPs



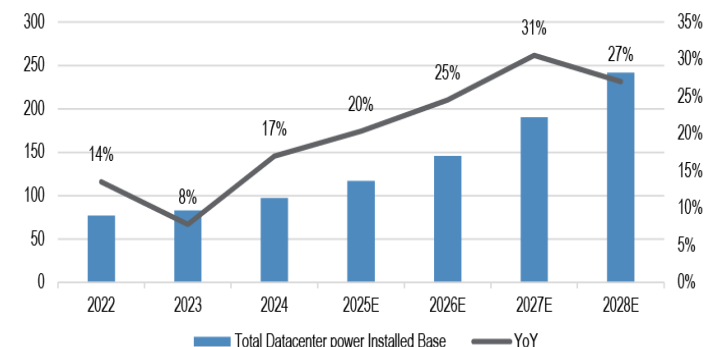
Source: 650 Group and J.P. Morgan Hardware and Networking estimates

Annual Change in Data Center Capex from Top 4 U.S. CSPs



Source: 650 Group and J.P. Morgan Hardware and Networking estimates

Datacenter Power Installed Base and YoY Trend



Source: IEA and J.P. Morgan EE/MI estimates

2025 JPM EPR Conference Takeaways

Standout Meetings: AEE, WEC; Remain Positive on ETR Load Opportunity

Key Takeaways from JPM Natural Resources Conference

AEE	PRP's 1.2GW sales est by 2030 an underestimate, Fall 2026 IRP positions AEE to fold in more capex at 3Q, where a higher CAGR guide could again point to the high end
AEP	Large load momentum alive and well; expect 3Q capex refresh to capture upside projects (including \$10bn shadow); \$16bn cost offsets & PJM options in spotlight
AQN CN	Rate case cycle advancing; medium-term growth catalysts building
CMS	Familiar face, new CFO Sri Maddipati, seamlessly steps in; renewed data center contracting positivity; NorthStar changes discussed at 2Q; next IRP brings upside
DTE	And now we wait: Google approval decision, 3Q IRP, possible data center contract coming later this year; Vantage data center project behind schedule
ED	Capex upside still in play, but conversion remains in question; affordability narrative topical
ETR	Next stage of data center clustering involves wider footprint; upside levers include investment in 5GW of solar, accelerated resilience, faster sales ramp, more transmission
FE	Constructive tone outside PA; WV gen + FET continue to gain momentum; genco structure under consideration
MDU	Bakken East execution in spotlight ahead of anticipated 7c filing in 3Q26; Applied ESA adds incremental data center demand marker
NJR	Regulated utility anchors durable growth; robust tailwinds across the platform
OGE	Large load tariff filing strikes balance between new and legacy load; capex catalyst path in focus into next plan refresh
PCG	Encouraged on SB 254 reform, while looking to avoid 'penny-wise, pound-foolish' tradeoffs
PEG	Seeing improved NJ footing ahead of BPU report, with eyes on multiple ways to drive constructive ratemaking; palpable PJM frustration; no change in new nuclear views
POR	Expressed hope for holdco settlement; positive feedback to-date on WA utility deal; affordability narrative strengthened by tariff + load; 2026 guide reaffirmed
PPL	Genco JV shines as key near-term watch item; look for upward movement in KY gen opportunities and potential for next CPCN; PAPUC approval de-risks utility
SWX	Great Basin optionality adds to fully regulated, high-visibility growth platform
WEC	Summer planning cycle underway to inform 3Q plan update, Point Beach in focus; 4-5GW opportunity for MSFT/Vantage sites; petitioning to reopen VLC tariff
XEL	CO electric settlement, MN electric verbal decision help de-risk regulatory exposure; elevated wildfire risk does not beget elevated utility risk; more RFP visibility incoming

See our accompanying note [here](#).

Source: J.P. Morgan

Energy Transition: Advanced Nuclear, LDES, CCS in Focus

Fireside Chat Series Examines Energy Transition Pathways

Last Mile Decarbonization

- **We see renewables deployment steadily increasing over time**
- **However, to achieve the last mile of decarbonization, we see the need for new technology, with these four pathways standing out**
 - **Long Duration Energy Storage (LDES)**
 - **Advanced Nuclear**
 - **Carbon Capture & Sequestration**
 - **Hydrogen**

Advanced Nuclear Call Takeaways

- **BWXT**: positioned as a components and fuel supplier for advanced reactors
- **CATF**: cost competitiveness and licensing requirements improvements in focus
- **DOE**: enthusiasm stands out in existing reactors, advanced tech, and spent fuel initiatives
- **DOE Loan Programs**: analysis includes repurposing old coal/natgas for SMR use
- **GE Hitachi**: notable BWRX momentum with OPG, TVA, SaskPower, and int'l partnerships
- **MIT**: first of a kind issues for large-scale projects are increasingly well understood
- **NuScale**: target efficiencies in mature LWR tech and asset light business model
- **Oklo**: pushing a new paradigm to advanced reactor licensing versus traditional process
- **OPG**: focused on US/Canada industry coordination, driving alliance around BWRX option
- **PEG**: positive nuke sentiment continues to grow amid global energy security focus
- **Pillsbury**: reg improvements will mitigate advanced tech, life extension licensing friction
- **TerraPower**: Natrium design includes a storage system for ramp between 100-500MW
- **Terrestrial**: industrial applications in focus as a zero carbon cogen solution
- **Westinghouse**: micro reactor targets off-grid/remote community and military solutions
- **X-energy**: proprietary TRISO-X fuel underpins safety and flexible operational use cases

CCS Call Takeaways

- **LADNR**: path to Class VI primacy; permitting timeline 1.0-1.5 years; Class V acceleration
- **Navigator**: Heartland project targets ethanol; 30-40mtpa addressable emissions market
- **Summit**: >\$1bn of new equity in the bank; IA ROW developments, eyeing 3Q24 ISD
- **Wolf**: leveraging Canadian expertise + ADM relationship to service US Midwest ethanol
- **Carbon Solutions**: building software to help optimize CCS system development
- **DOE**: optimal CO2 storage; current projects; future incentives
- **CATF**: CCS in industrial projects, power sector; CCS economics and the need for hub approach
- **IEA**: CCUS plays vital role in meeting Net Zero goals
- **BRK/GaffneyCline**: CCS market from activation to scale; US geology; pipeline conversion
- **Clean Energy**: driving global adoption of CCUS
- **Rice**: TX CCS advantage; RRC primacy for TX; thoughts on Houston hub
- **University of Houston**: hub model and Houston as the first mover
- **8 Rivers**: developing a circular economy; prefers carbon tax vs credit; Allam-Fetvedt Cycle
- **Dr. Julio Friedmann**: advised on Class VI permitting; need policy to lead the way for growth

LDES Call Takeaways

- **LDES Council**: indicated TES can optimize industrial heat consumption and facilitate the reuse of waste heat, estimating cost optimized pathway of \$540bn in savings with 2-8 TW
- **Brenmiller Energy**: bGen technology uses crushed rocks to store heat, which can be used for industrial processes or converted into steam to generate electricity
- **Malta**: tried-and-true molten salt-based storage manufacturer, targets utility-scale applications with base storage of 100 MW with 10-200 hr duration (10-12hr most popular)
- **DOE Energy Storage**: highlighted potential for smaller scale, repeatable solutions (e.g., flow batteries, electro-chemical storage) can address a bigger TAM
- **Hydrostor**: CAES technology's advantage lies within ability to offer diurnal storage to use cases beyond lithium ion's capabilities or do not have access to pumped hydro storage

J.P.Morgan

Energy Transition: Long Duration Energy Storage

Fireside Chat Series Underscores Growing Need for LDES to Revolve Renewable Intermittency

Top Takeaways from 5 Expert Fireside Chats Regarding Long Duration Energy Storage

- Long duration energy storage momentum surges with increasing use cases to accompany renewable intermittency and emissions mandates
 - Within the LDES world, the DOE highlighted significant unanswered questions across which LDES technologies will win out in the 10+ and 24+ hour duration markets
 - Commercializing the 20-50 hour duration market stands out as important, such as working with PSPS
 - IRA tax incentives cover at least 50% of the DOE Earthshot Initiative's expected timeline to get to 90% reduction in 10+ hour storage costs by 2040 (from a 2020 lithium ion baseline), even after accounting for recent supply chain constraints
- LDES technologies and use cases stretch far and wide, with each developer seeking to offer a distinct competitive advantage to the peer group
 - **Hydrostor** targets a 10-hour compressed air storage system, costing ~\$3,000/kW, with a \$50/kW increase for every additional hour (up to 48 hrs)
 - **Brenmiller's** TES bGen Unit offers a 4-24 hour duration range, with long term plans to commercialize 10-500 MW units (current projects 1-4 MW)
 - **Malta** primarily competes within 10-12 hour duration markets (up to 200 hrs), offering base unit storage of 100 MW and can flex up as needed
- See our accompanying notes [Series I](#) and [Series II](#)

Individual Call Takeaways

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Coverage

DUK, NEE, POR, SO, XEL

- DUK is targeting 1,600 MW of battery storage by 2029
 - Including EnerVenue (nickel hydrogen), Honeywell (flow battery), Eos Z3 (zinc), lithium ion, and lithium iron phosphate technology
- NEE is targeting 5,100-6,800 MW of incremental energy storage by 2023-2026
- POR is targeting 475 MW of battery storage by 2025
- SO is partnering with Form Energy to deploy a 15 MW battery by 2026
- XEL is partnering with Form Energy to deploy two iron-air battery systems at retired coal plants

Energy Transition: Advanced Nuclear

Fireside Chat Series Underscores Technological Evolution, Changing Sentiment

Top Takeaways from 15 Expert Fireside Chats Regarding Advanced Nuclear Energy

- Changing nuclear sentiment evident with widespread acknowledgement over the long-term need for nuclear to meet clean energy goals
 - Dr. Ralph Izzo of PSEG discussed the importance of energy security especially with nuclear producing 40% of carbon free energy in the US
 - CATF highlighted nuclear's recognition in the EU draft green taxonomy as important to facilitating the next generation of new nuclear technologies
 - An acknowledgement of the need to utilize all available tools to decarbonize the global energy economy
- SMRs build in passive safety, allowing widespread deployment, and seek efficiencies with standardized designs. SMRs come in all shapes and sizes!
 - NuScale 77 MW light water reactor offers tangible benefits versus traditional nuclear plants and is first to receive NRC design approval
 - Oklo 1.5 MW fast reactor can recycle nuclear waste for fuel, improving economics even further as input costs lower
 - X-Energy Xe-100, is an 80MWe/200MWt high temperature gas cooled reactor requiring 1/10th of the components as a traditional nuclear plant
 - Westinghouse 5MW eVinci micro reactor, which is transportable, scalable, and employs a passive cooling system in a compact unit
 - Dr. Huff of the DOE looks for a grid-connected SMR by 2029 with smaller scale reactors potentially even sooner
- See our accompanying notes [Series I](#), [Series II](#), [Series III](#), [Series IV](#), [Series V](#), and [Series VI](#)

Individual Call Takeaways

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Favorable Nuke Sentiment Feeds into Our PEG OW

PEG Thesis:

- Revamped business features a low risk T&D utility
 - Potential meaningful upside leverage to offshore wind transmission
- Attractive valuation relative to premium peers
- Nuclear PTCs in the Inflation Reduction Act supports PEG's 3.8GW nuclear fleet
 - Significantly de-risk non-regulated cash flows
 - Provides greater line of sight to support versus NJ's current three year ZEC authorization

Energy Transition: Carbon Capture & Sequestration

Fireside Chat Series Underscores Strong Growth Opportunities Ahead

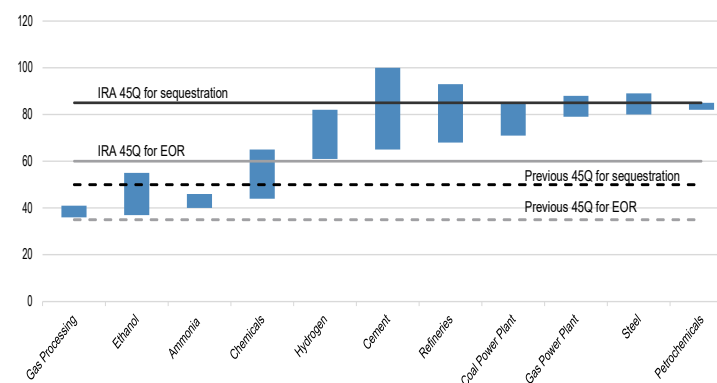
Top Takeaways from 17 Expert Fireside Chats Regarding Carbon Capture Sequestration and Utilization

- CCS appears to present the clearest path to decarbonize certain industrial processes, particularly ethanol & ammonia and hard to abate steel & cement
 - Near term large private projects in focus: Summit Pipeline, Navigator Heartland Greenway Project, Wolf
- We continue to believe CCS could present a compelling opportunity for midstreamers to participate in energy transition, potentially enabling new growth opportunities that leverage existing assets and capabilities
 - \$85/ton 45Q tax credit improves CCS economics and incentivizes widespread deployment beyond near purity CO₂ sources such as ethanol to industrial emitters
- Coordination of action across capture-transport-storage and streamlining permitting and regulatory oversight to provide greater timeline clarity would also benefit optimal CCUS deployment
 - Class VI well permitting timelines remain a key point of uncertainty, with IL, ND, SD, and NE standing out as more favorable states
 - Current process of gaining pipeline right of way key to ethanol CO₂ pipeline development
- See our accompanying notes [Takeaways from first 12 chats](#), [CCUS Series IV Takeaways](#), and [2023 CCUS Series with Summit, Navigator, & Wolf](#)

Individual Call Takeaways

- **LADNR:** path to Class VI primacy; permitting timeline 1.0-1.5 years; Class V acceleration
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CCUS Economics



Source: J.P. Morgan

- **GaffneyCline CCUS cost estimates:** capture (\$20-\$100s/ton), transport (\$5-\$25/ton), storage (\$3-\$25/ton)
- **Navigator CCS rough capex breakdown:** 30% capture, 60% pipeline, and 10% sequestration

Surveying the Data Center Landscape

Covered Company Survey Highlights Sprawling Data Center Activity

Data Center Survey Overview

- We present company responses to a set of four data center related questions for an overview of demand forecasts, tariff structure/economics, regulatory and political outlooks, and differentiated benefits
- Operators continue to look across the country in search for lowest rates and speed to market
- Debate over bill impacts and sustainability drives the regulatory outlook, with regulatory risk stemming from long lead times
- Transmission constraints loom large

Data Center Standouts in the Utility Space

- **NEE** and **AES** more exposed to data center growth with their unregulated renewables businesses
 - **NEE** well poised to leverage its scale, data, technology and supply chain to deliver PPAs for data center developers
- Continued development in northern VA (**D**) has spilled over into nearby regions including the Carolinas (**DUK**), GA (**SO**), OH (**AEP**), and MD (**FE**)
- Notable data center initiatives also materializing in MS (**ETR**), WI (**WEC**), CA (**PCG**), and OR (**POR**)
- Overall, we see expeditious access to low-cost and reliable power screening as the leading factor behind data center siting decisions
 - Southeast and Midwest appear as comparatively attractive regions to accommodate these needs

Data Centers One-Liners for Our Coverage (Utilities)

Company	Key Takeaways
AEE	MO and IL pursue data centers, with MO offering competitive large C&I rates, regulatory support, and available infrastructure
AEP	Commercial load growth forecasts of 4.1-7.6% YoY through 2026 largely driven by data centers, most prevalent in IN, OH, and TX
AES	Excess capacity from OH transmission line resonates as prime mega data center area; OH D42 tariff and IN proposed riders stand out
CMS	While MI remains attractive for data center, manufacturing load tends to attract greater number of jobs and state investment
CNP	Estimated +20-25MW five-year data center load growth, but higher expected C&I growth with associated residential opportunities
D	~ 7.8GW pipeline of data centers looking to join the grid over the next 3-6 years, necessary investments support rate base growth
DTE	Any data center additions would be upside to plan; MI has abundance of electricity and water to service data center growth
DUK	Data centers will drive an estimated 25% of new commercial load (up from 10% now), utilizing traditional C&I cost-of-service rates
ED	NY's clean energy transition carries more material significance than data center development
EIX	Given CA's electrification focus, EIX has not been seeing the same data center demand growth as other areas in the country
EMA	EMA couples a high growth service, favorable TEC service territory with longer term decarbonization opportunities at NS Power
ETR	Experience with specific rates for large industrial customers, supportive regulatory environments, and low carbon opportunities
EXC	2019 IL data center law exempts large data centers from sales tax; EXC largely decoupled, not seeing materiality in load forecasts
FE	LT industrial load forecast of 62.4 M MWhs in 2028E (vs 55.5 M MWhs in 2023A) includes data center growth (primarily MD)
FTS	FTS continues to receive IA and MI interest, driven by state incentives, quick , and access to renewable generation
NEE	As the largest renewables developer in the U.S., NEE remains well positioned to offer renewable PPAs to data center developers
NI	IN offers sales and use tax exemptions for eligible investments; could see upside to load growth forecasts with NI's upcoming IRP
NWE	With legislative reforms and more generation, NWE could provide a low temperature, high renewables, low cost service territory
PCG	Northern CA remains top data center hub in US, with 175MW contributing to PG&E's system peak in 2023
PEG	PSEG nuclear offers carbon-free base load; NJ NJEDA could decide to attract new industry via Artificial Island port
POR	Data centers and high tech manufacturing have supported notable industrial load growth, aided by proximity to subsea cables
SO	Use of one-time fees to limit cost increases and scale of 24x7 load growth emphasized amid regulatory concerns over bill impacts
SRE	While Oncor does not invest in generation assets, data center consumption can positively impact earnings and create headroom
WEC	Reliability, access to water, and limited exposure to extreme weather attracted Microsoft investment without tax incentives
XEL	Data centers expected to drive C&I load growth; CO may pass MN-style tax incentives; low carbon generation attracts developers

Source: J.P. Morgan

G.R.E.A.M. (Green Rules Everything Around Me)

The ESG Investment Opportunity and Utilities

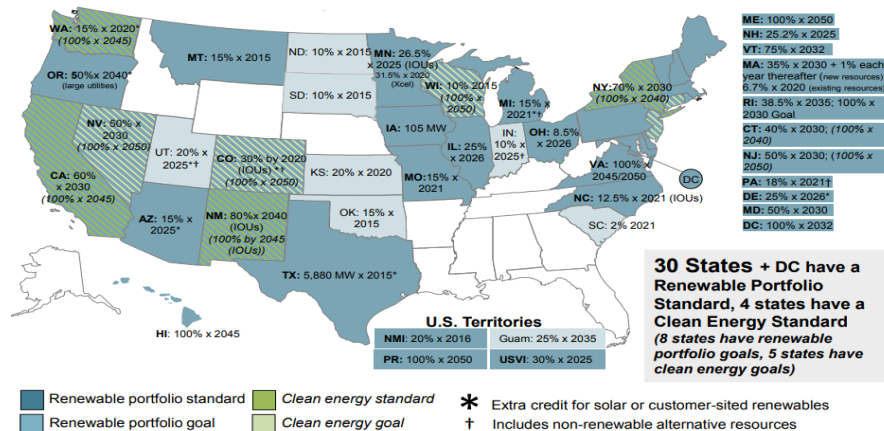
Green Investment Opportunity

- US utilities can represent an investment in environmental change
 - Ideally positioned to capitalize on state environmental policies that continue increasing and accelerating renewable energy initiatives
- More investment leads to growing rate base and utility returns
 - Augmented by replacing fossil fuel commodity pass-through expenses
- Economics of renewable energy increasingly competitive
 - Federal climate legislation could help improve the economics for renewables/EVs
- C&I customer desire to decrease their carbon footprint feed into renewable generation demand

An Investment in Clean Energy Future

- ESG influenced funds an increasing % of incremental AUM
 - One of the primary drivers of premium utility valuations
- ESG Sustainable Strategies:
 - Negative/exclusionary strategy – excludes certain sectors, companies or practices (ex fossil fuels)
 - ESG integration – focuses on systematic/explicit consideration of ESG in investment analysis and decisions
- Impact Investing represents an interesting opportunity for utilities
 - Could benefit rate of change stories
- Represent the best expression of decarbonization for investors given that their market cap dwarfs the Alt Energy space, with value flowing down to the end asset owner
- Identifying names with “Green” attributes or gaining these attributes can drive alpha

State Policy Supports Energy Transition



Zero emissions credit subsidies

- IL, NY, and NJ provide subsidy payments to at-risk nuclear facilities based on carbon-free energy attributes

Energy efficiency standards

- Require utilities to reduce customer usage
- Almost always receive compensation for lost load

Electric vehicles

- Pilot programs for charging stations and vehicle buyers receive subsidies

Battery investments

- Federal and state efforts encourage investment, including an important recent policy investigation by the FERC

Source: Database of State Incentives of Renewable Energy.

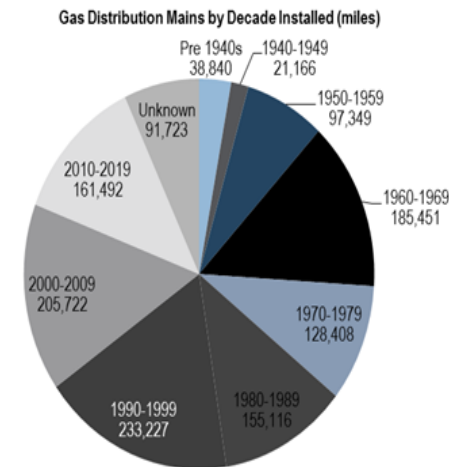
Natural Gas in Focus

NatGas Infrastructure – Location Matters

Negative Gas Sentiment Offsets Above Avg Growth

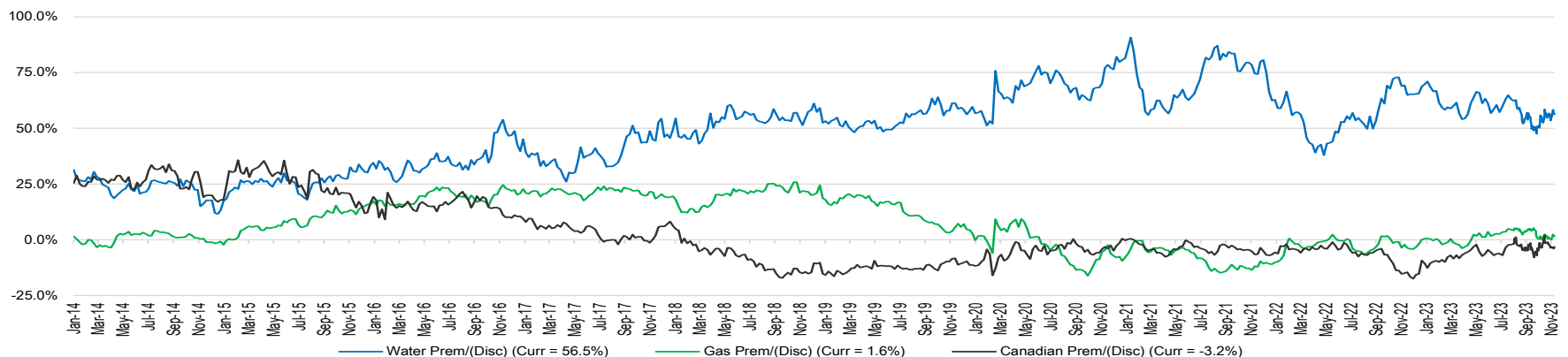
- Modernizing underpins solid growth
 - Half nation's distribution pipe originating before 1970 or unknown
 - Increasingly stringent safety and integrity standards target aggressively replacing aging bare steel & cast iron pipes
 - Low natgas prices have incentivized switching from heating oil & propane
- Select pushback with certain localities favoring electrification that leverages renewables rather than future gas consumption
 - San Jose and Oakland residential and commercial gas bans
 - MA DPU investigating LDC role in achieving emissions targets
 - NY forcefully denied pipeline solutions
 - Investor fears of diminished growth outlooks,
 - if not potential stranded asset risk
- Electric & gas multi-utility valuations show resilience
- Hydrogen and RNG can decarbonize the gas system

Gas Distribution Mains by Decade Installed



Source: U.S. Department of Transportation, Pipeline and Hazardous Materials Safety Administration.

Historical Gas LDC Premium to Regulated Utilities Has Eroded



Source: Bloomberg Finance L.P., J.P. Morgan.

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Valuation Framework

A Combination of Quantitative and Qualitative Factors

- **Our approach to individual company valuation and process for considering P/E premiums/discounts to peer average multiples:**
- **Regulatory backdrop leveraging RRA rankings:**
 - More constructive ratemaking typically featuring riders and trackers to mitigate lag, future test years or post-test year adjustments, multi-year plans or formula rates, above-average ROEs, and consistent regulatory rulings, among other factors
 - Overall, favorable regulation ultimately translates into an ability to consistently earn returns at or close to authorized levels
 - Our analysis quantitatively compares our utility coverage universe based on weighted rate base avg jurisdictional exposure, leveraging RRA
- **DPS growth within the context of EPS payout ratio:**
 - We use a 5% average 3-year dividend CAGR as our anchor, with relative premium/discounts assigned accordingly
 - We adjust dividend growth credit based on payout ratio relative to 65%, as well as change in payout ratio over the corresponding 3-year period
- **Financial flexibility:**
 - Credit metric headroom provides meaningful flexibility to limit equity needs, cushion against unexpected events, y seek out M&A
 - We believe mid-BBB sets the minimum threshold for investor comfort on HoldCo leverage. We assign relative premiums/discounts vs BBB
- **Management track record:**
 - Valuation premiums for consistency
 - Even the perception of aggressive assumptions or capex plan “white space” can both weighing on multiples
- **Growth from regulated operations:**
 - Given regulated utilities’ strong operating fundamentals & investor preferences, we see greater value in growing rate base vs non-reg biz
 - Historically management track record and firm contracting could mitigate these deltas, but not apparent in the current market
 - Do not see the catalyst to reverse this dynamic going forward, which feeds into our favorable bias towards regulated growth

Utility Coverage Universe

Comps Set

Company	Ticker	Rating	Lead Analyst	Mkt Cap (\$ MM)	EV (\$ MM)	Price Target	Price
Regulated							
Ameren Corporation	AEE	OW	Jeremy Tonet	\$29,036	\$50,961	\$133.00	\$104.90
American Electric Power	AEP	N	Jeremy Tonet	\$67,141	\$121,174	\$140.00	\$123.33
CenterPoint Energy, Inc.	CNP	N	Jeremy Tonet	\$25,534	\$48,428	\$46.00	\$39.11
CMS Energy Corporation	CMS	OW	Jeremy Tonet	\$21,057	\$40,862	\$81.00	\$67.15
DTE Energy Company	DTE	N	Jeremy Tonet	\$27,580	\$55,341	\$156.00	\$132.58
Duke Energy Corp.	DUK	N	Jeremy Tonet	\$93,148	\$187,766	\$137.00	\$119.42
Consolidated Edison Inc.	ED	UW	Jeremy Tonet	\$39,372	\$65,908	\$112.00	\$106.46
Edison International	EIX	N	Aidan Kelly	\$21,550	\$66,495	\$61.00	\$56.00
Eversource Energy	ES	UW	Jeremy Tonet	\$25,810	\$53,651	\$73.00	\$68.52
Entergy Corp.	ETR	OW	Jeremy Tonet	\$49,150	\$80,235	\$137.00	\$105.33
Exelon Corp.	EXC	N	Jeremy Tonet	\$44,160	\$94,412	\$47.00	\$43.16
FirstEnergy	FE	N	Jeremy Tonet	\$26,683	\$56,966	\$52.00	\$46.13
NiSource Inc.	NI	OW	Eli Jossen	\$19,859	\$39,486	\$52.00	\$41.41
PG&E Corp.	PCG	OW	Aidan Kelly	\$30,393	\$95,706	\$19.00	\$13.80
PPL Corporation	PPL	OW	Jeremy Tonet	\$25,647	\$45,638	\$44.00	\$34.08
Public Service Enterprise Group	PEG	N	Jeremy Tonet	\$36,081	\$60,567	\$81.00	\$72.39
The Southern Company	SO	N	Jeremy Tonet	\$98,356	\$175,231	\$101.00	\$87.17
WEC Energy Group	WEC	N	Jeremy Tonet	\$34,322	\$57,690	\$121.00	\$105.33
Xcel Energy	XEL	OW	Jeremy Tonet	\$47,160	\$85,473	\$97.00	\$75.50
Mean				\$40,107	\$77,999		
Median				\$30,393	\$60,567		
Mean (ex EIX, PCG & POR)				\$36,400	\$67,482		
Median (ex EIX, PCG & POR)				\$28,308	\$56,154		
SMID							
MDU Resources Group Inc	MDU	OW	Aidan Kelly	\$4,039	\$6,570	\$27.00	\$19.20
NorthWestern Energy	NWE	N	Aidan Kelly	\$4,259	\$7,833	\$73.00	\$69.24
OGE Energy Corp	OGE	OW	Aidan Kelly	\$9,616	\$15,459	\$54.00	\$46.55
Portland General Electric	POR	N	Aidan Kelly	\$5,673	\$10,849	\$53.00	\$48.24
Mean				\$5,897	\$10,178		
Median				\$4,966	\$9,341		
Diversified							
Sempra	SRE	OW	Aidan Kelly	\$54,496	\$98,423	\$102.00	\$83.34
Mean (Regulated & Diversified)				\$40,827	\$79,021		
Median (Regulated & Diversified)				\$32,357	\$63,237		
Mean (ex EIX & PCG)				\$43,467	\$80,740		
Median (ex EIX & PCG)				\$36,081	\$60,567		
LDCs							
Atmos Energy	ATO	N	Eli Jossen	\$27,577	\$37,476	\$180.00	\$163.19
New Jersey Resources	NJR	OW	Eli Jossen	\$5,394	\$9,231	\$63.00	\$53.16
Spire Inc	SR	N	Eli Jossen	\$4,741	\$11,984	\$85.00	\$80.20
Southwest Gas Holdings Inc.	SWX	OW	Eli Jossen	\$6,231	\$9,471	\$102.00	\$86.11
Mean				\$10,986	\$17,041		
Median				\$5,813	\$10,727		
Canadians							
Algonquin Power & Utilities Corp.	AQN	N	Eli Jossen	\$5,658	\$15,776	\$10.00	\$7.35
Emera Inc.	EMA	N	Eli Jossen	\$20,949	\$44,092	\$75.00	\$68.19
Fortis Inc.	FTS	UW	Eli Jossen	\$38,499	\$78,232	\$77.00	\$75.36
Mean				\$21,702	\$46,033		
Median				\$20,949	\$44,092		
Water							
American Water Works Company, Inc.	AWK	N	Jeremy Tonet	\$27,363	\$43,220	\$147.00	\$137.69
Mean				\$14,993	\$23,815		
Median				\$14,993	\$23,815		

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